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BDA assignment (Databricks)

The following five labs needs to be performed with screenshots for every lab

1. **Spark SQL Basics**

Import a CSV dataset (e.g., sales or student records) into Databricks Community Edition and run simple SQL queries: `SELECT`, `GROUP BY`, `ORDER BY`.

- *Learning outcome*: Transition from Hive SQL to Spark SQL in Databricks.

2. **Delta Lake Time Travel**

Create a small Delta table, update a few rows, and query older versions using `VERSION AS OF`.

- *Learning outcome*: Understand ACID transactions and schema evolution compared to Hadoop HDFS.

3. **Streaming Data Simulation**

Use a JSON file stream (simulated IoT sensor data) and process it with Spark Structured Streaming.

- *Learning outcome*: Compare with Flume/Kafka ingestion, but in a modern streaming pipeline.

4. **DataFrame Transformations**

Load a dataset into a Spark DataFrame, perform filtering, joins, and aggregations, then visualize results with built-in charts.

- *Learning outcome*: Reinforce Spark programming concepts beyond Sqoop/Hive.

5. **Genie Agent Demo**

Register a small dataset in Unity Catalog (e.g., sales or movie ratings), configure a Genie Agent, and ask natural-language queries like “Show top 5 products by revenue.”

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- *Learning outcome*: Experience conversational BI and see how SQL is auto-generated.

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Theory:

A Lakehouse Architecture is a modern hybrid data platform that combines the low-cost, scalable storage of a data lake with the reliability, governance, and performance of a data warehouse. It allows organizations to store raw structured, semi-structured, and unstructured data while still supporting ACID transactions, schema enforcement, and advanced analytics.

🔑 Key Characteristics of Lakehouse Architecture

- Unified Storage: Uses cloud object storage (e.g., AWS S3, Azure Blob, GCP Storage) for raw data.
- Open Table Format: Layers like Delta Lake or Apache Iceberg add metadata, schema enforcement, and versioning.
- ACID Transactions: Ensures reliable reads/writes, unlike traditional data lakes.
- Schema Evolution: Pipelines adapt without breaking when data changes.
- Governance: Unified security and cataloging across structured and unstructured data.
- Performance: Warehouse-grade query speed with engines like Photon or Presto.

How It Works

1. Storage Layer: Raw data stored in low-cost cloud object storage.
2. Metadata Layer: Tracks schema, versions, and transactions (Delta Lake/Iceberg).
3. Compute Layer: Engines like Spark, Databricks SQL, or BI tools query the data.
4. Medallion Architecture: Organizes data into Bronze (raw), Silver (cleaned), and Gold (aggregated) layers for progressive quality.

Problems It Solves

- Eliminates Silos: No need to maintain separate data lakes and warehouses.
- Reduces Duplication: Same dataset serves raw exploration and BI analytics.
- Improves Freshness: Data is available for real-time streaming and batch pipelines.
- Supports AI/ML: Unified platform for BI, predictive analytics, and generative AI.

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Comparison: Data Lake vs Warehouse vs Lakehouse

Feature	Data Lake	Data Warehouse	Lakehouse
Data Types	**Raw structured, semi-structured, unstructured**	**Structured only**	**All types unified**
Cost	**Low**	**High**	**Low (storage) + optimized compute**
Transactions	**No ACID**	**ACID supported**	**ACID supported**
Governance	**Weak**	**Strong**	**Unified**
Use Cases	**Raw storage, ML training**	**BI, reporting**	**BI + ML + streaming**

✅ Why It Matters for You (Teaching Context)

Since your students already know Hadoop, Flume, and Sqoop:

- Data Lakehouse (Databricks) shows them the evolution beyond Hadoop.
- They'll see how Delta Lake improves over HDFS with ACID and time travel.
- They'll understand how Genie Agents and BI tools query the same unified dataset.
- This prepares them for industry demand, where Databricks and Lakehouse skills are highly valued.

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STEPS:

Lab 1: Spark SQL Basics in Databricks

1. Setup Databricks Community Edition

- Sign up at Databricks Community Edition (community.cloud.databricks.com in Bing).
- Create a new **Notebook** (Python or SQL).

2. Download Dataset

Use a simple public CSV dataset. For example:

- **Student Performance Dataset**: student_scores.csv (raw.githubusercontent.com in Bing)
- **Retail Sales Dataset**: retail_sales.csv (raw.githubusercontent.com in Bing)

👉 These are small, free datasets suitable for classroom labs.

3. Upload Dataset to Databricks

- In the left sidebar, go to **Data** → **Add Data** → **Upload File**.
- Upload `student_scores.csv` or `retail_sales.csv`.
- Databricks will store it in `/FileStore/tables/`.

4. Load Dataset into Spark DataFrame

python

```
# Load CSV into DataFrame
df = spark.read.csv("/FileStore/tables/student_scores.csv", header=True,
inferSchema=True)

# Show first 5 rows
df.show(5)
```

5. Create Temporary SQL View

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python
```
# Register DataFrame as SQL view
df.createOrReplaceTempView("students")
```

6. Run Spark SQL Queries

sql
```
-- Select all records
SELECT * FROM students;

-- Count students by gender
SELECT gender, COUNT(*) AS total_students
FROM students
GROUP BY gender;

-- Average math score by parental education
SELECT parental_level_of_education, AVG(math_score) AS avg_math
FROM students
GROUP BY parental_level_of_education
ORDER BY avg_math DESC;

-- Top 5 students by reading score
SELECT student_id, reading_score
FROM students
ORDER BY reading_score DESC
LIMIT 5;
```

7. Visualize Results

- After running a SQL cell, click **+** → **Visualization**.
- Choose **Bar chart** or **Pie chart** to visualize grouped results.
- Example: Average math score by parental education → Bar chart.

🎯 Learning Outcomes

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- Students see how **Spark SQL** queries look similar to Hive SQL but run on Databricks.
- They learn to **import CSVs, create views, run queries, and visualize results**.
- This lab bridges their Hadoop background with modern **lakehouse workflows**.

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Lab 2: Delta Lake Time Travel

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Create a Sample Dataset

python

```
# Create sample data
data = [
    (1, "Alice", 85),
    (2, "Bob", 90),
    (3, "Charlie", 78)
]

columns = ["student_id", "name", "score"]

df = spark.createDataFrame(data, columns)
df.show()
```

3. Write Data to Delta Table

python

```
# Save as Delta table
df.write.format("delta").mode("overwrite").save("/tmp/delta/students")

# Read back
delta_df = spark.read.format("delta").load("/tmp/delta/students")
delta_df.show()
```

4. Update Data (Versioning)

python

```
from delta.tables import DeltaTable
```

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```
# Load Delta table
delta_table = DeltaTable.forPath(spark, "/tmp/delta/students")

# Update score for Alice
delta_table.update(
    condition = "name = 'Alice'",
    set = { "score": "95" }
)

# Show updated data
delta_table.toDF().show()
```

5. Add New Record

python

```
new_data = [(4, "David", 88)]
new_df = spark.createDataFrame(new_data, columns)

new_df.write.format("delta").mode("append").save("/tmp/delta/students")

# Show current table
spark.read.format("delta").load("/tmp/delta/students").show()
```

6. Query Older Versions (Time Travel)

python

```
# Check history
delta_table.history().show()

# Read version 0 (original data)
spark.read.format("delta").option("versionAsOf", 0).load("/tmp/delta/students").show()

# Read version 1 (after Alice update)
spark.read.format("delta").option("versionAsOf", 1).load("/tmp/delta/students").show()
```

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7. Learning Outcomes

- Students see how **Delta Lake** supports ACID transactions.
- They learn **time travel**: querying older versions of data.
- They understand why Delta Lake is superior to plain HDFS (which lacks versioning and schema enforcement).

This lab clearly shows the **evolution from Hadoop HDFS → Delta Lake**.

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Lab 3: Streaming Data Simulation in Databricks

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Create a Simulated Streaming Source

We'll simulate streaming data by writing JSON files into a directory.

python

```
import time
import json

# Create a directory for streaming data
dbutils.fs.mkdirs("/tmp/streaming_data")

# Simulate writing JSON events
for i in range(5):
    event = {"event_id": i, "sensor_value": i*10, "timestamp": str(time.time())}
    path = f"/tmp/streaming_data/event_{i}.json"
    dbutils.fs.put(path, json.dumps(event), overwrite=True)
```

3. Read Stream with Structured Streaming

python

```
# Read streaming data
stream_df = (
    spark.readStream
    .format("json")
    .schema("event_id INT, sensor_value INT, timestamp STRING")
    .load("/tmp/streaming_data")
)

# Display stream in real-time
display(stream_df)
```

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4. Process Streaming Data

python

```
# Aggregate sensor values
agg_df = (
    stream_df.groupBy("event_id")
    .agg({"sensor_value": "avg"})
    .withColumnRenamed("avg(sensor_value)", "avg_value")
)

# Write results to memory table
query = (
    agg_df.writeStream
    .outputMode("complete")
    .format("memory")
    .queryName("sensor_avg")
    .start()
)
```

5. Query Results

sql

```
-- Run this in a SQL cell
SELECT * FROM sensor_avg;
```

6. Visualize Results

- Use **bar chart visualization** to show average sensor values by event_id.
- Students can see results update as new JSON files are added.

7. Learning Outcomes

- Students understand how **Structured Streaming** replaces Flume/Kafka ingestion pipelines.

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- They see **real-time aggregation** in Databricks.
- They learn how streaming data can be queried continuously, unlike batch Hadoop jobs.

This lab demonstrates the **shift from batch Hadoop ingestion → real-time streaming in Databricks**.

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Lab 4: DataFrame Transformations in Databricks

1. Setup Notebook

- Open Databricks Community Edition.
- Create a new notebook (Python).

2. Load Sample Dataset

We'll use a small CSV dataset (students or retail sales). Upload it via **Data** → **Add Data** → **Upload File**.

Example: `student_scores.csv` with columns: `student_id, name, gender, math_score, reading_score, writing_score`.

python
```
# Load CSV into DataFrame
df = spark.read.csv("/FileStore/tables/student_scores.csv", header=True, inferSchema=True)
df.show(5)
```

3. Filter Rows

python
```
# Filter students with math_score > 80
high_math = df.filter(df.math_score > 80)
high_math.show()
```

4. Select Specific Columns

python
```
# Select name and reading_score
df.select("name", "reading_score").show(5)
```

5. Group and Aggregate

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python
```
# Average scores by gender
avg_scores = df.groupBy("gender").agg(
    {"math_score": "avg", "reading_score": "avg", "writing_score": "avg"}
)
avg_scores.show()
```

6. Join Two DataFrames

python
```
# Create another DataFrame with student hobbies
hobbies = [
    (1, "Football"),
    (2, "Chess"),
    (3, "Music"),
    (4, "Dance")
]
hobby_df = spark.createDataFrame(hobbies, ["student_id", "hobby"])

# Join with main DataFrame
joined_df = df.join(hobby_df, "student_id", "left")
joined_df.show()
```

7. Sort Results

python
```
# Top 5 students by writing score
df.orderBy(df.writing_score.desc()).show(5)
```

8. Visualize Results

- After running aggregations, click **+** → **Visualization**.
- Example: Bar chart of average math score by gender.

🎯 Learning Outcomes

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- Students practice **filtering, selecting, grouping, joining, and sorting** with Spark DataFrames.
- They see how transformations are more flexible than Hive queries.
- They learn how to combine multiple datasets seamlessly.

This lab shows the **power of Spark DataFrames compared to Hadoop's batch-oriented tools**.

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Lab 5: Genie Agent Demo

1. Prerequisites

- Databricks Community Edition account.
- A small dataset uploaded to **Unity Catalog** (e.g., `student_scores.csv` or `retail_sales.csv`).
- A **SQL Warehouse** (Serverless or Pro) — required for Genie Agents.

2. Upload Dataset

- Go to **Data** → **Add Data** → **Upload File**.
- Upload `student_scores.csv` (columns: `student_id, name, gender, math_score, reading_score, writing_score`).
- Register the dataset in **Unity Catalog** under a schema (e.g., `default.students`).

3. Create Genie Agent

1. In Databricks workspace, go to **Genie Agents** (under AI/BI section).
2. Click **Create Agent**.
3. Configure:
    - **Name**: `Student Performance Agent`
    - **Datasets**: Add `default.students` table.
    - **SQL Warehouse**: Select your running warehouse.
    - **Description**: “Agent to query student performance data.”

4. Ask Natural-Language Questions

Students can now interact with the agent:

Basic Query
Code
Show all student records.

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-

Aggregation
Code
What is the average math score by gender?

-

Top Results
Code
Show top 5 students by reading score.

-

Conditional Query
Code
List students who scored above 80 in writing.

-

5. View SQL Translation

- Genie will generate SQL automatically, e.g.:

sql
```
SELECT gender, AVG(math_score) AS avg_math
FROM students
GROUP BY gender;
```

Students can compare Genie's SQL with their own manual queries.

6. Visualize Results

- Genie supports charts directly.
- Example: Bar chart of average math score by gender.

7. Learning Outcomes

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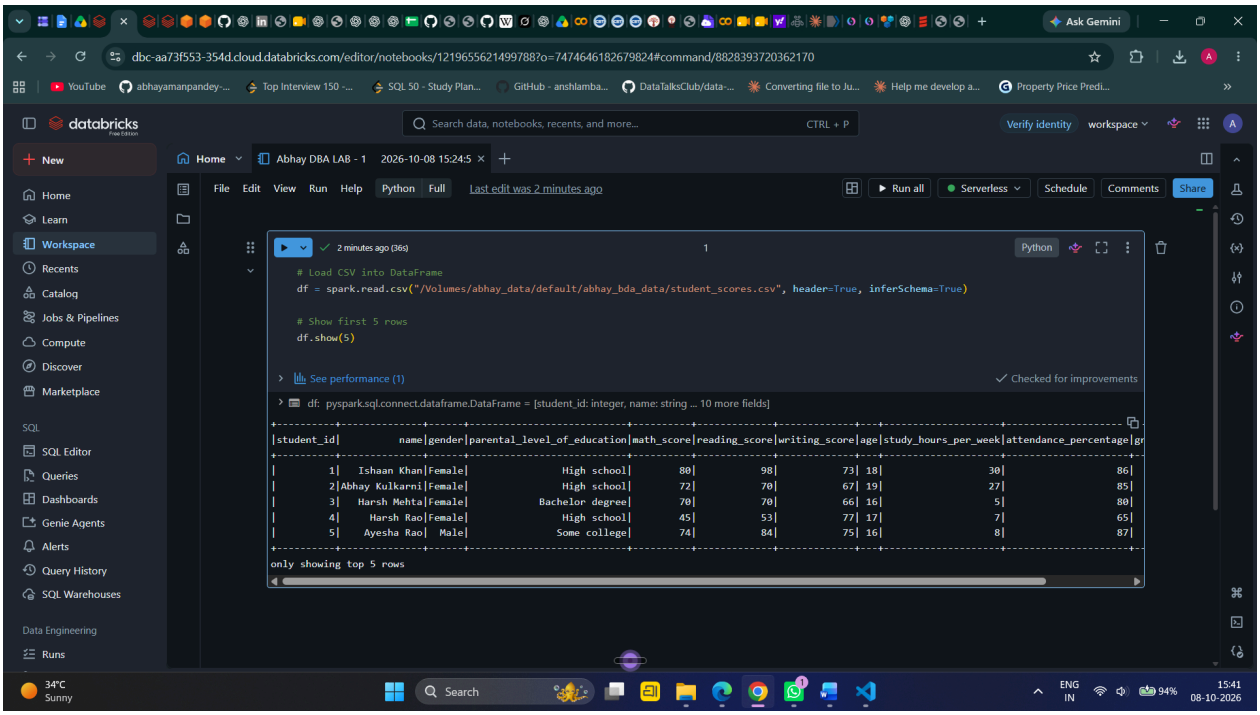
- Students experience **natural-language querying** in Databricks.
- They see how Genie auto-generates SQL, bridging technical and non-technical users.
- They understand how conversational BI integrates with **modern lakehouse workflows**.

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LAB 1 | Spark SQL Basics

OUTPUT / EXECUTION EVIDENCE

Figure 1.1 - lab01-output01.png



Source: GitHub repository / Output_Lab-1

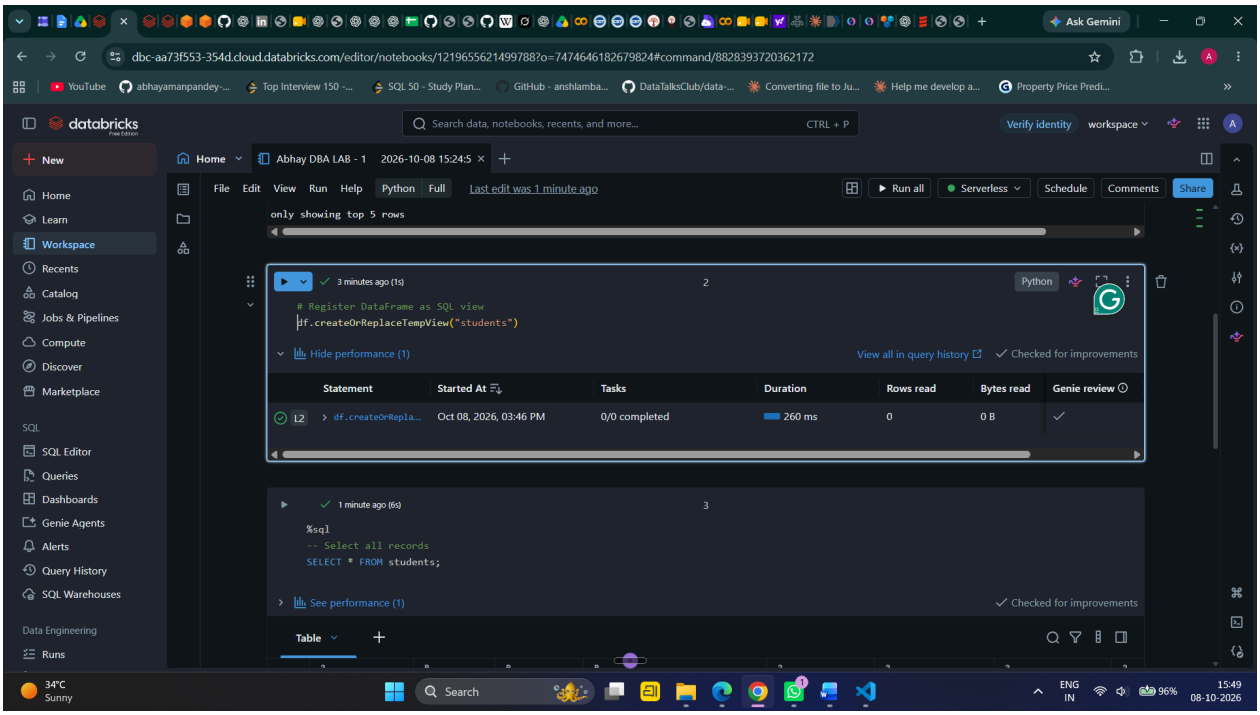
Execution screenshot 1 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

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Figure 1.2 - lab01-output02.png



Source: GitHub repository / Output_Lab-1

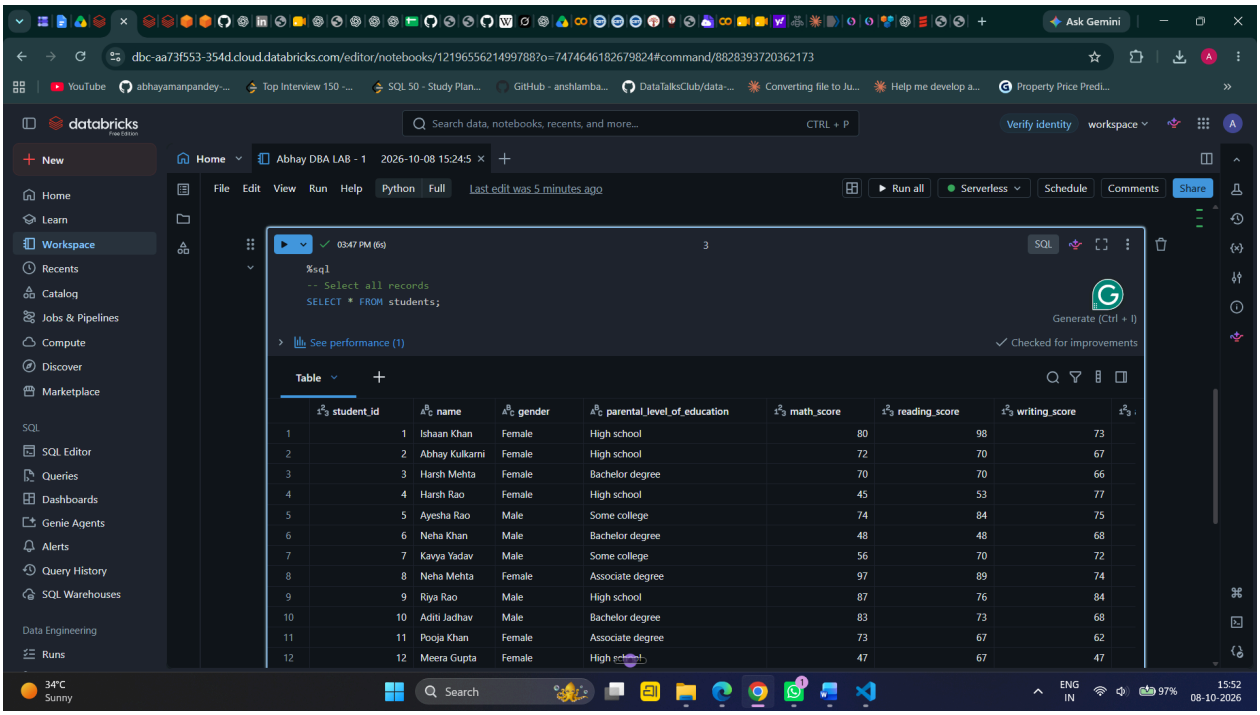
Execution screenshot 2 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

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Figure 1.3 - lab01-output03.png



Source: GitHub repository / Output_Lab-1

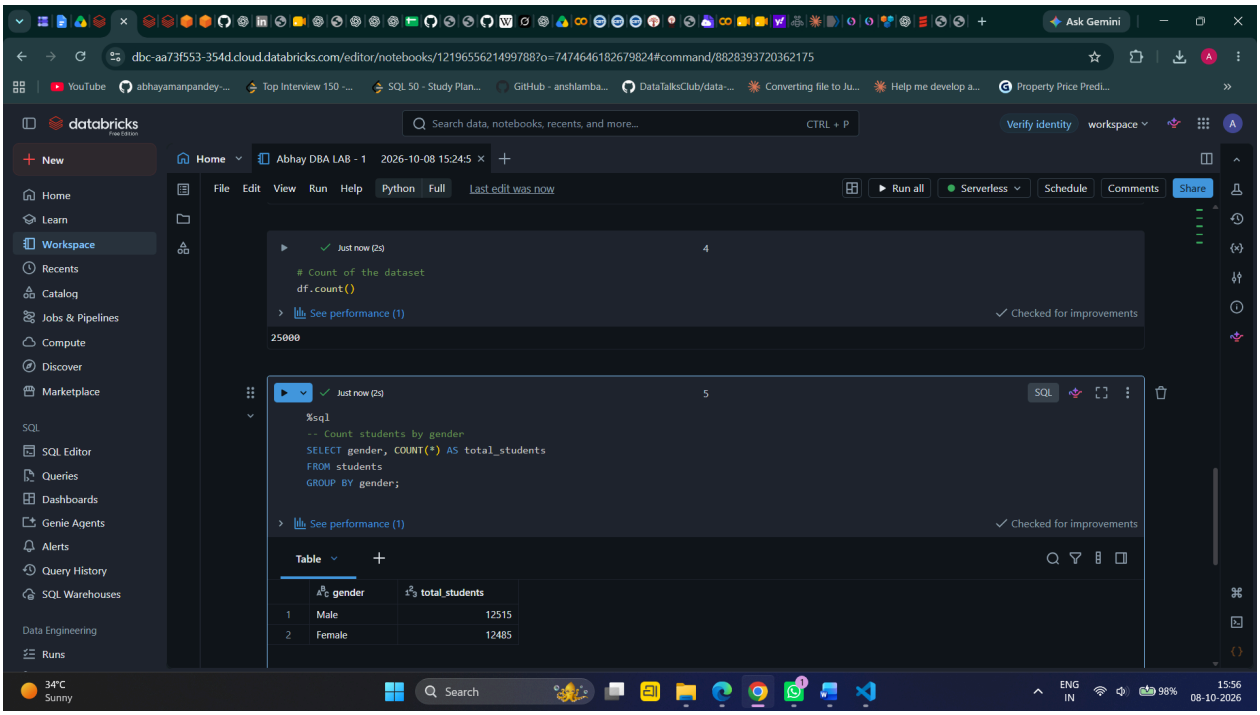
Execution screenshot 3 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

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Figure 1.4 - lab01-output04.png



Source: GitHub repository / Output_Lab-1

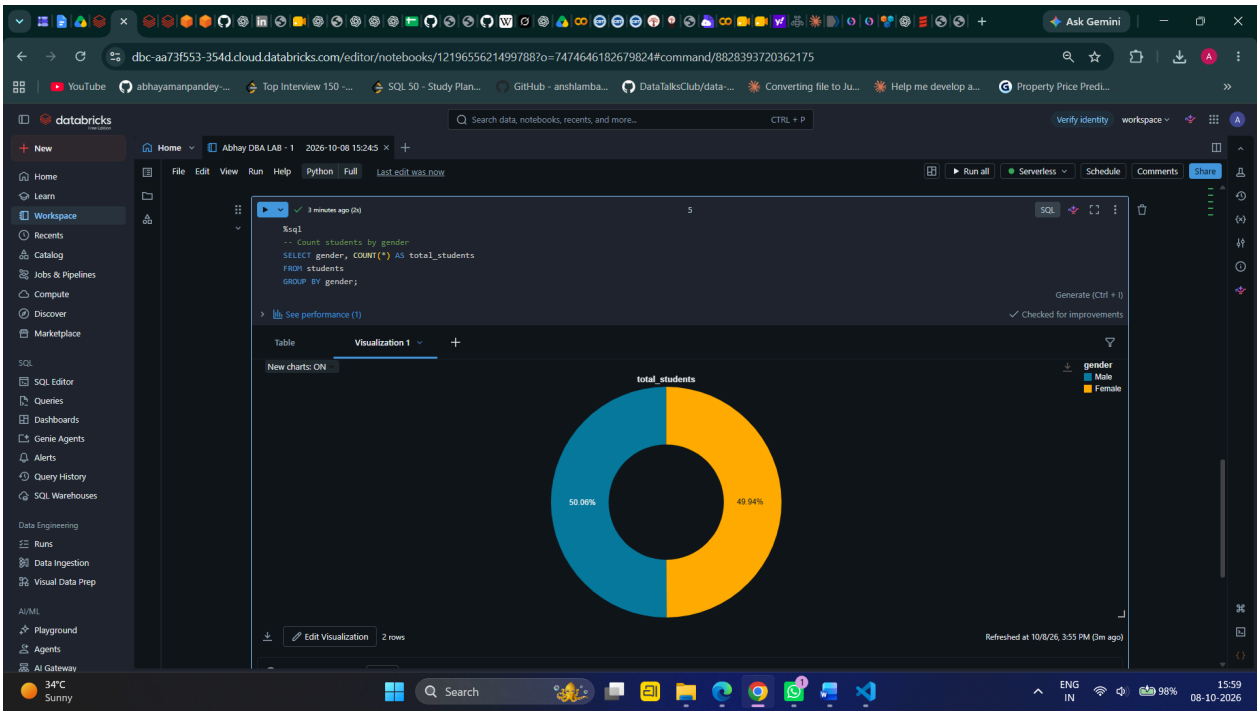
Execution screenshot 4 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

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Figure 1.5 - lab01-output05.png



Source: GitHub repository / Output_Lab-1

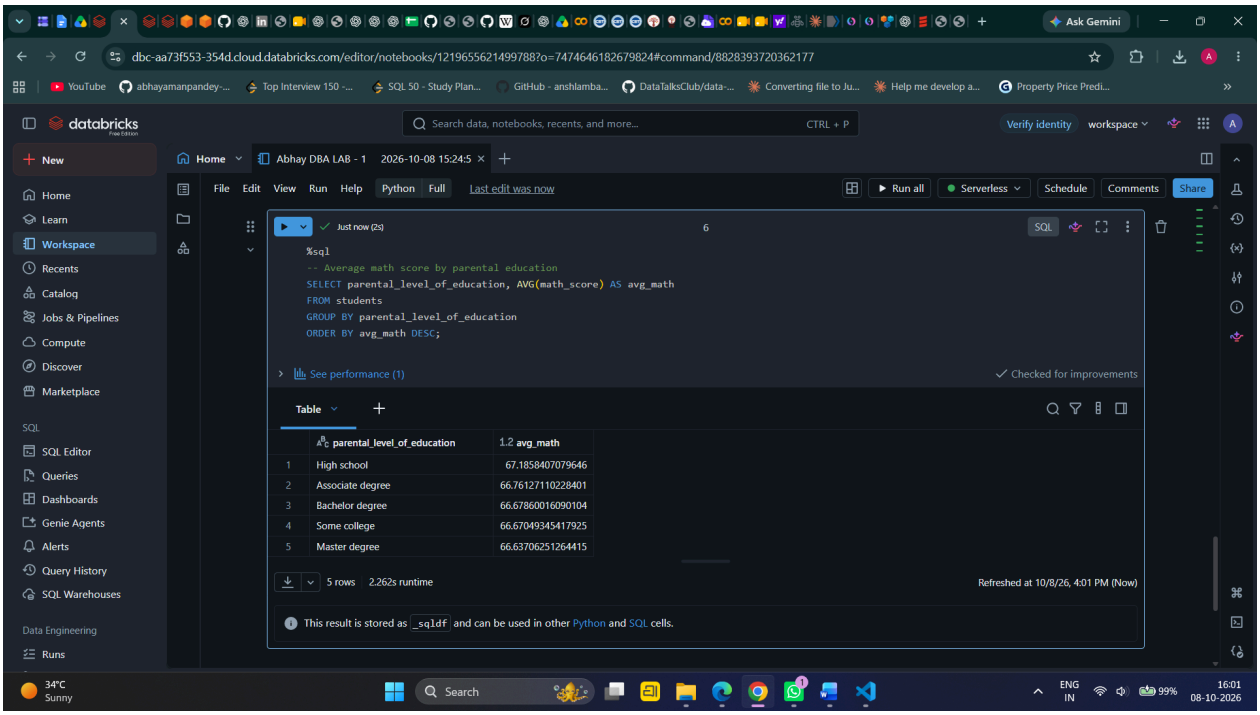
Execution screenshot 5 of 9 for Lab 1.

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OUTPUT / EXECUTION EVIDENCE

Figure 1.6 - lab01-output06.png



Source: GitHub repository / Output_Lab-1

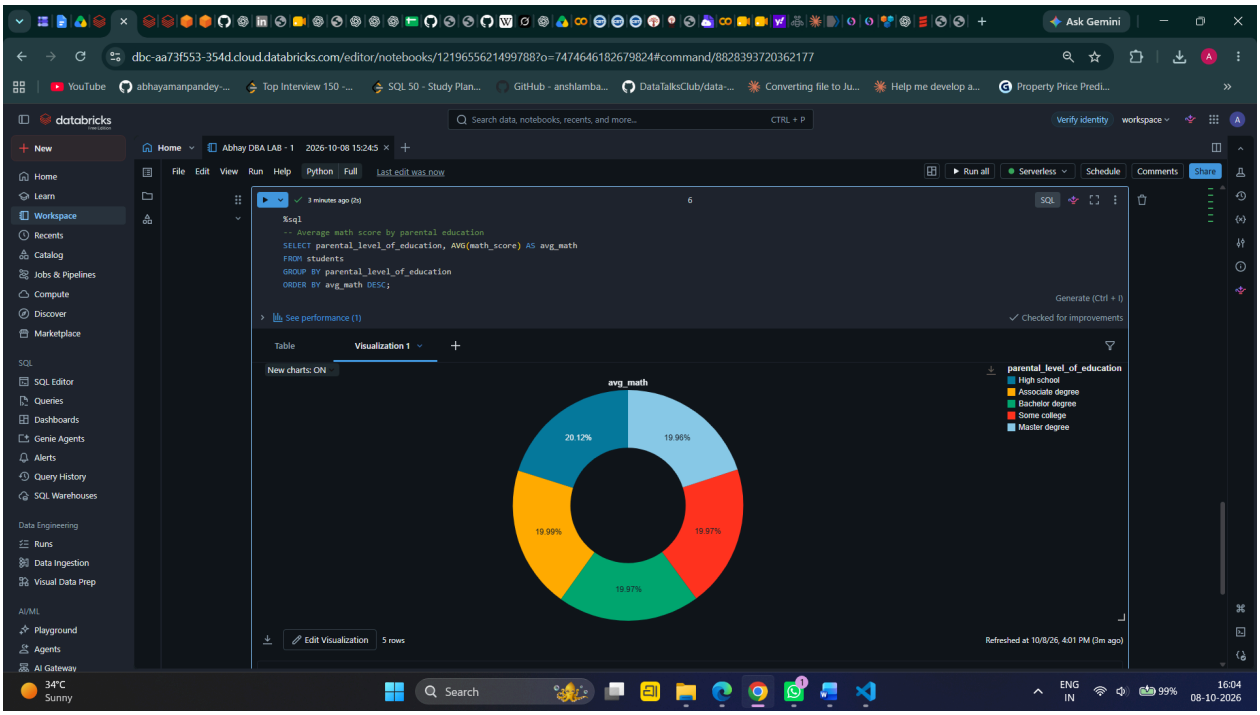
Execution screenshot 6 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

OUTPUT / EXECUTION EVIDENCE

Figure 1.7 - lab01-output07.png



Source: GitHub repository / Output_Lab-1

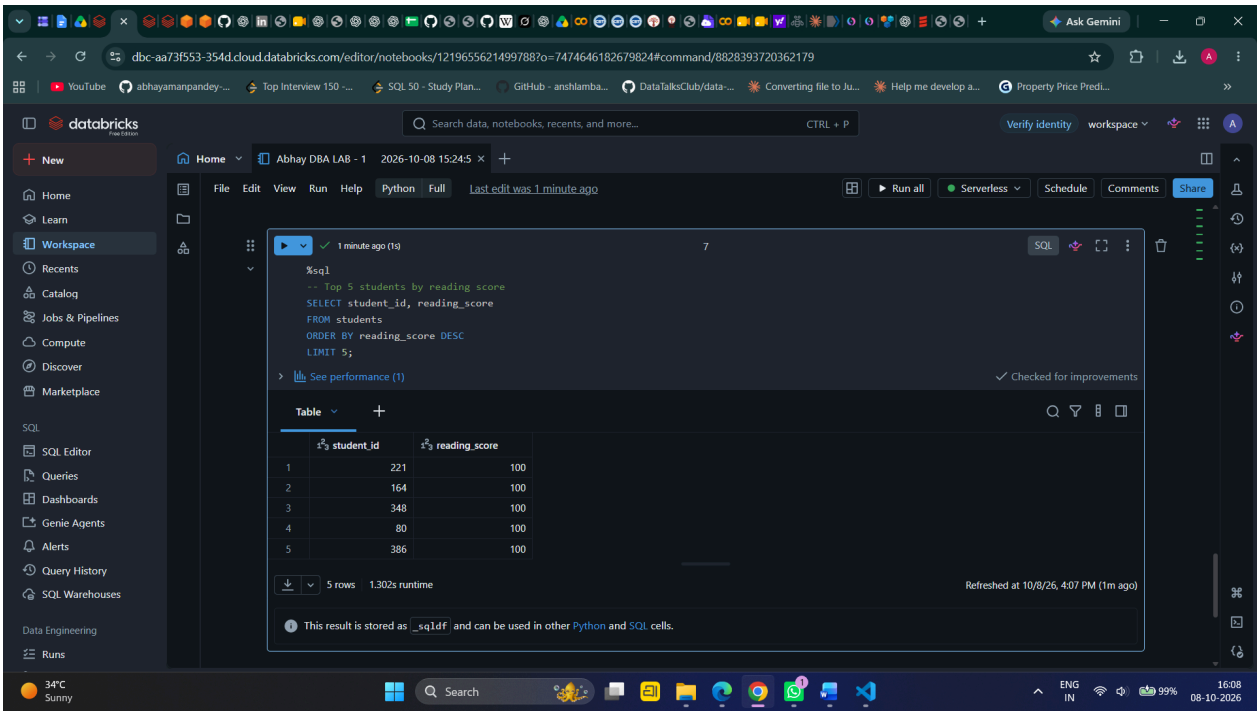
Execution screenshot 7 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

OUTPUT / EXECUTION EVIDENCE

Figure 1.8 - lab01-output08.png



Source: GitHub repository / Output_Lab-1

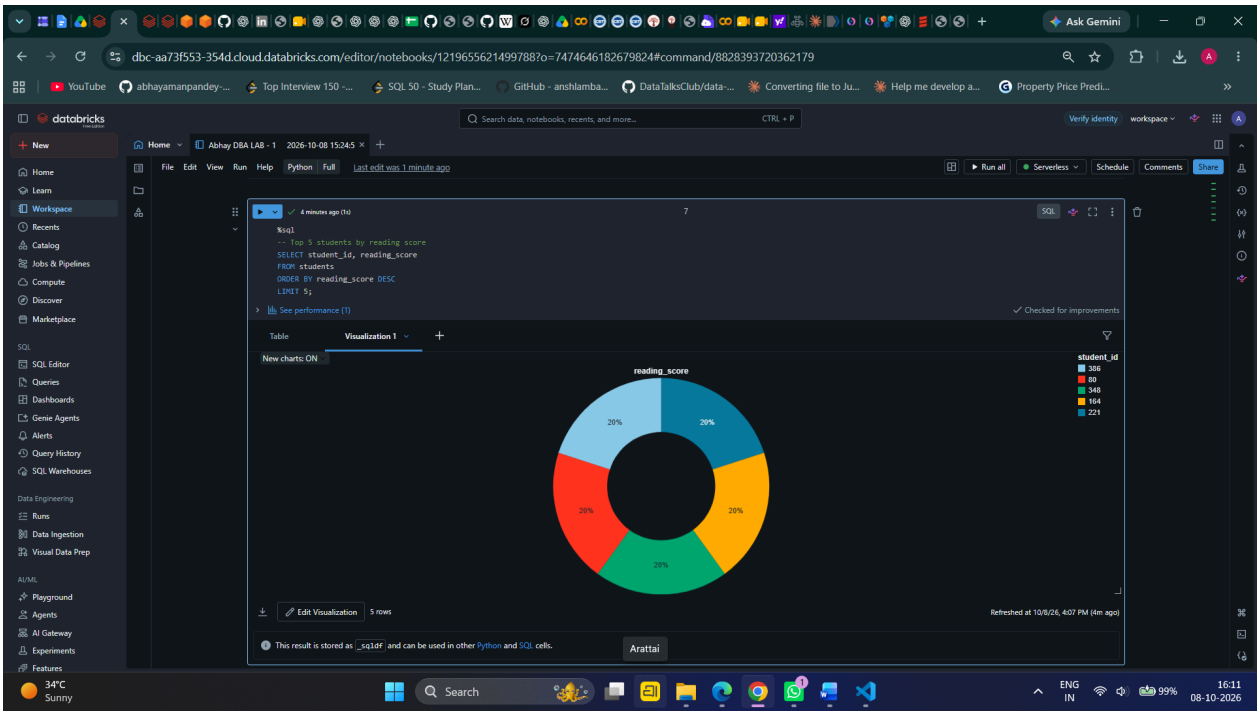
Execution screenshot 8 of 9 for Lab 1.

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LAB 1 | Spark SQL Basics

OUTPUT / EXECUTION EVIDENCE

Figure 1.9 - lab01-output09.png



Source: GitHub repository / Output_Lab-1

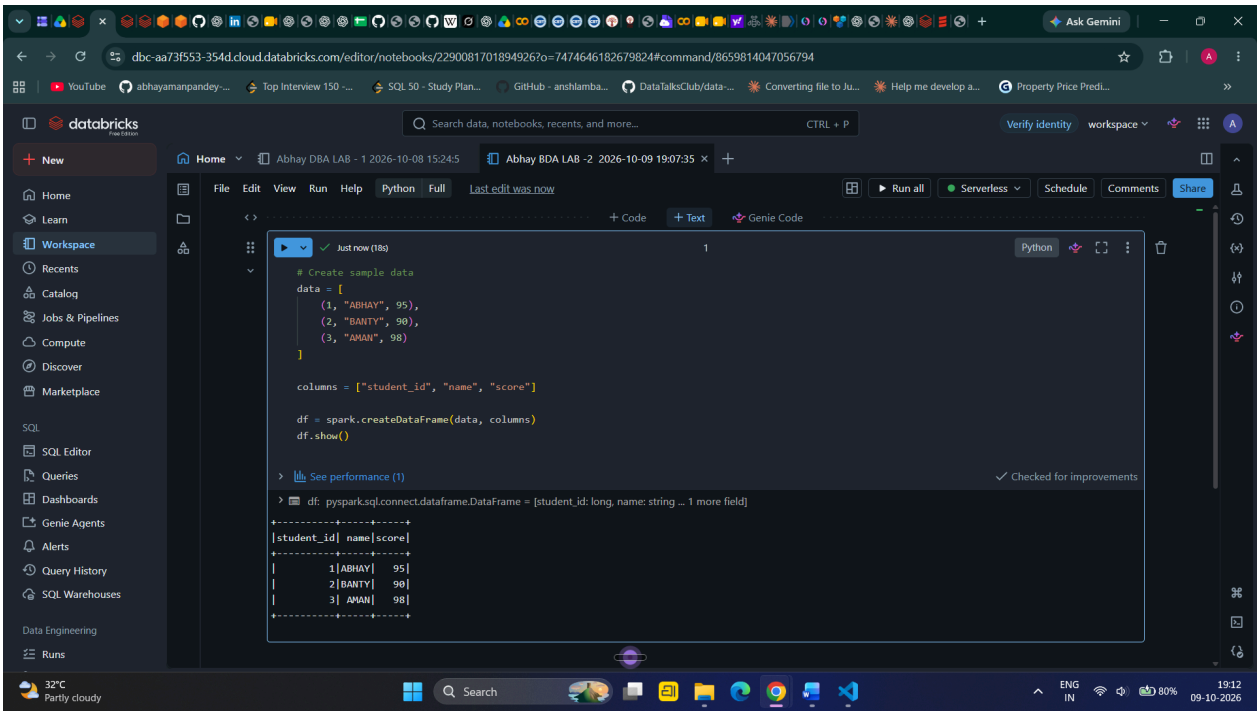
Execution screenshot 9 of 9 for Lab 1.

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LAB 2 | Delta Lake Time Travel

OUTPUT / EXECUTION EVIDENCE

Figure 2.1 - lab02-output01.png



Source: GitHub repository / Output_Lab-2

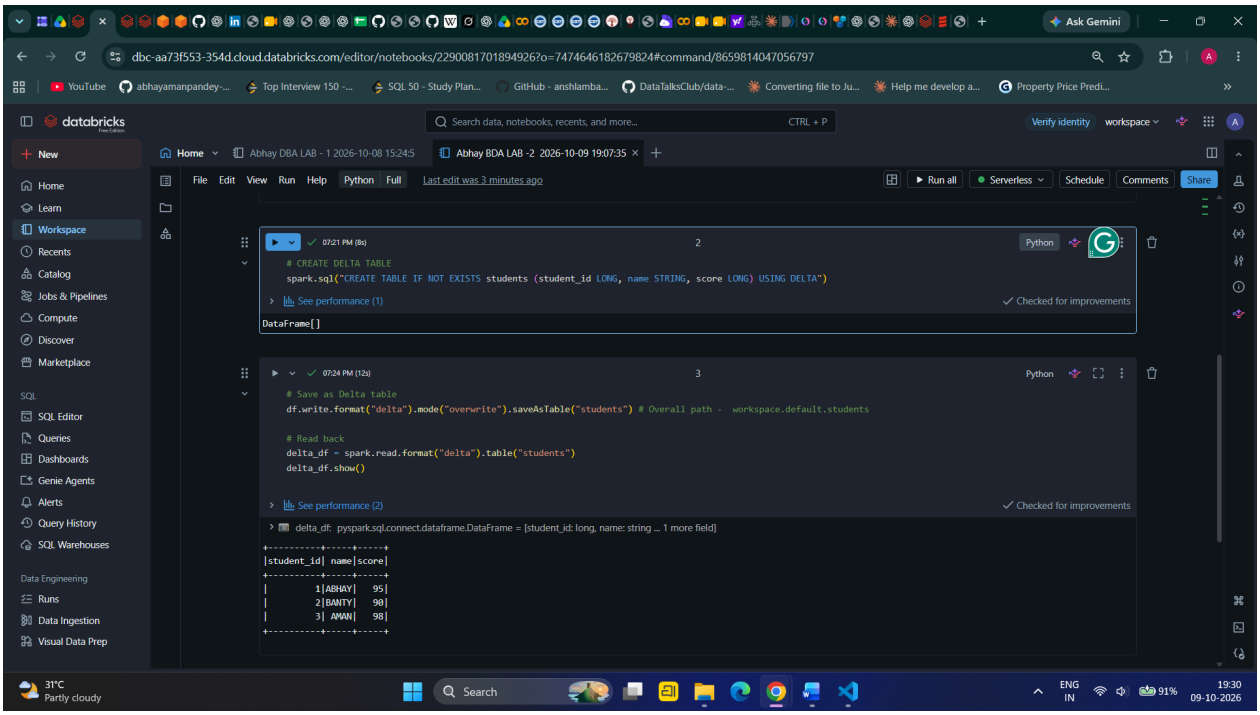
Execution screenshot 1 of 6 for Lab 2.

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LAB 2 | Delta Lake Time Travel

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Figure 2.2 - lab02-output02.png



Source: GitHub repository / Output_Lab-2

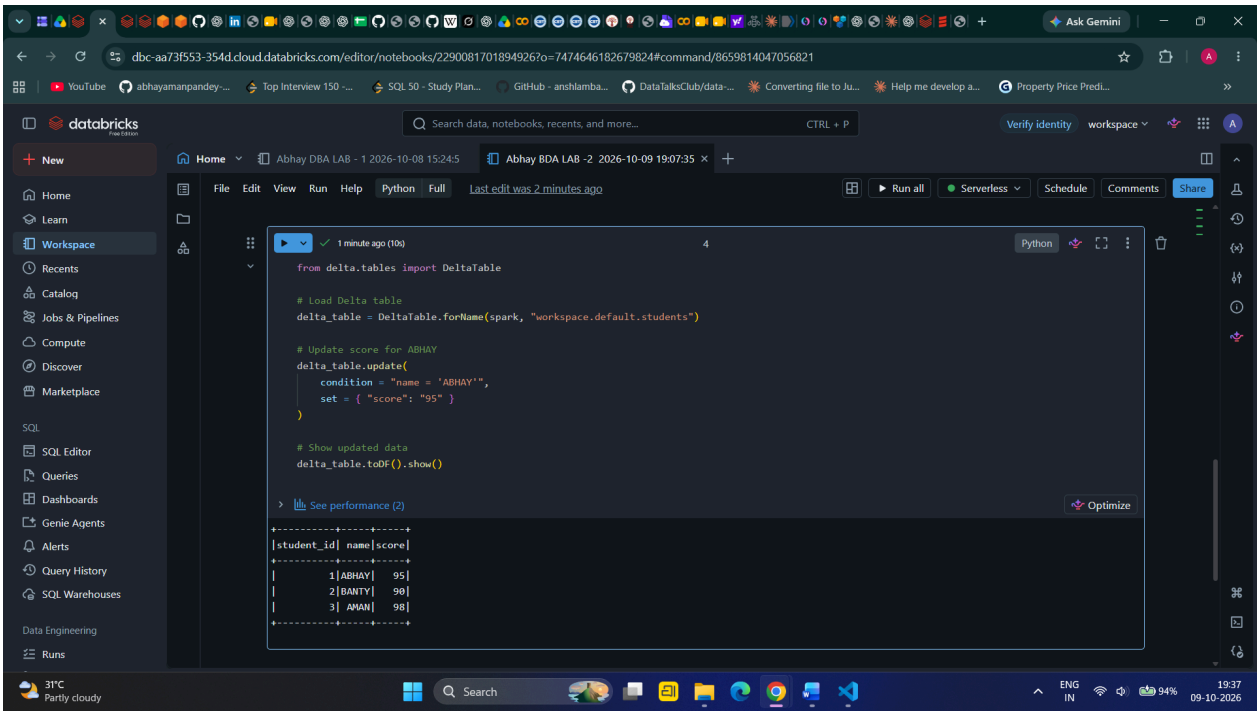
Execution screenshot 2 of 6 for Lab 2.

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LAB 2 | Delta Lake Time Travel

OUTPUT / EXECUTION EVIDENCE

Figure 2.3 - lab02-output03.png



Source: GitHub repository / Output_Lab-2

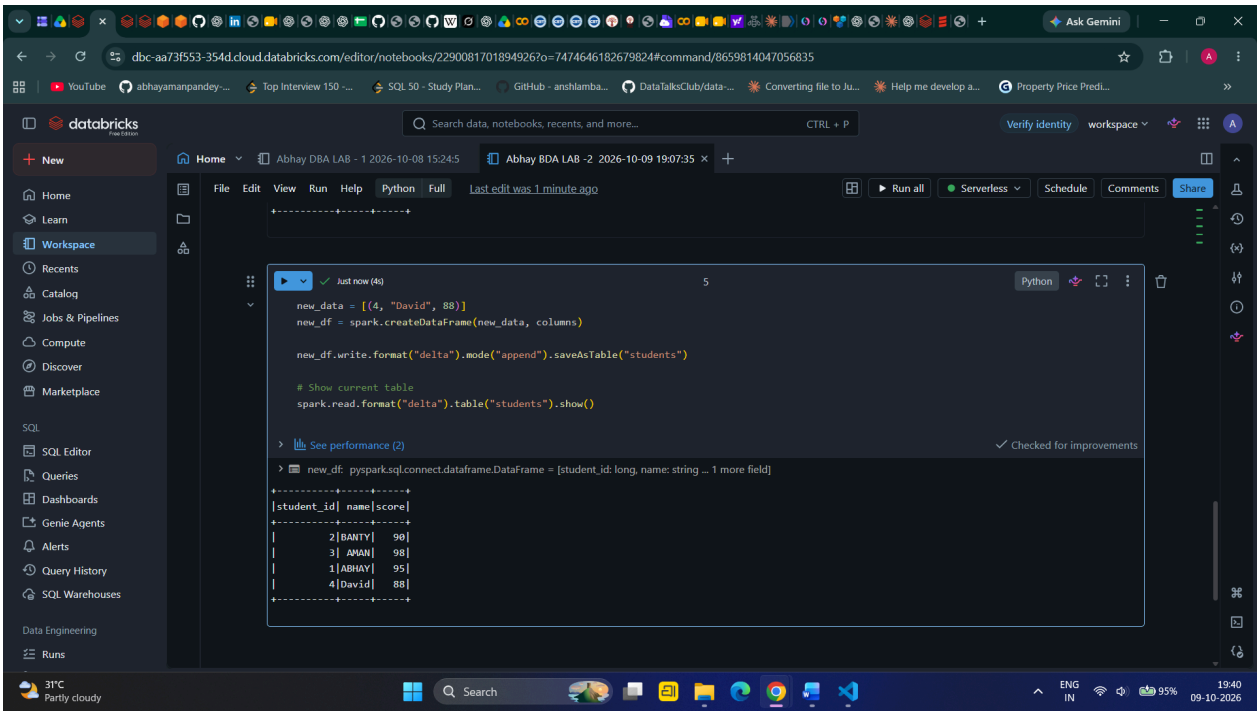
Execution screenshot 3 of 6 for Lab 2.

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LAB 2 | Delta Lake Time Travel

OUTPUT / EXECUTION EVIDENCE

Figure 2.4 - lab02-output04.png



Source: GitHub repository / Output_Lab-2

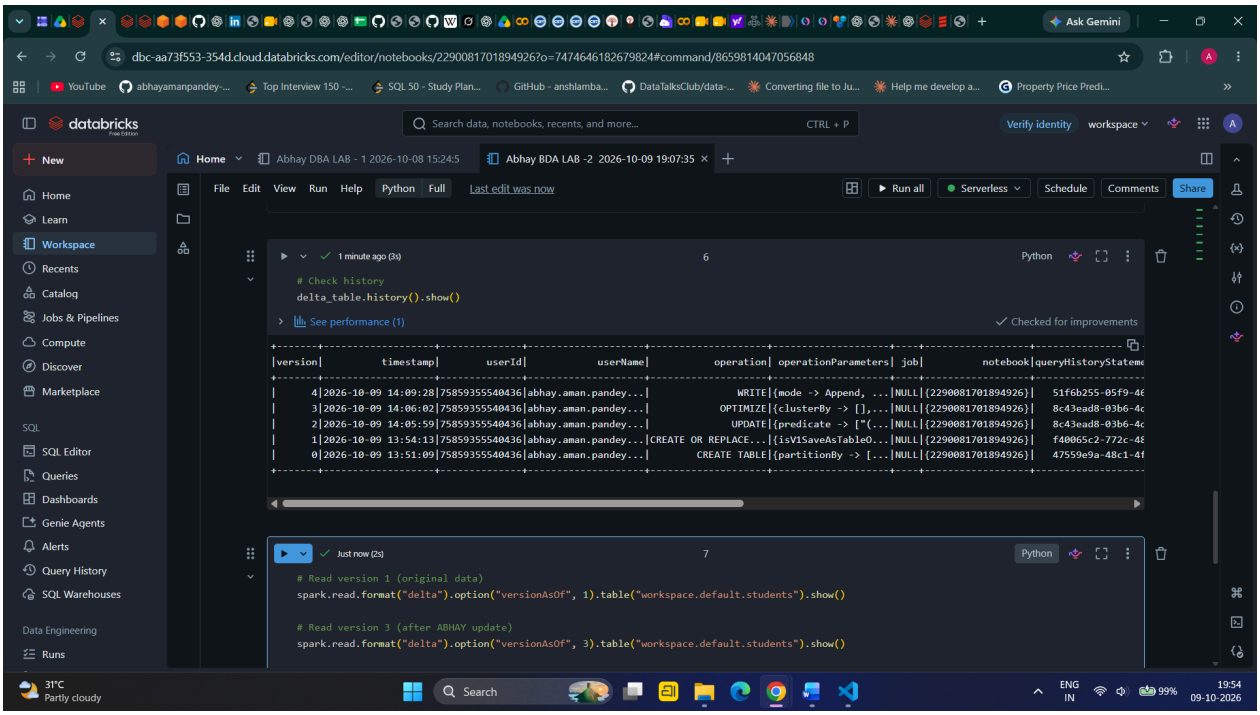
Execution screenshot 4 of 6 for Lab 2.

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LAB 2 | Delta Lake Time Travel

OUTPUT / EXECUTION EVIDENCE

Figure 2.5 - lab02-output05.png



Source: GitHub repository / Output_Lab-2

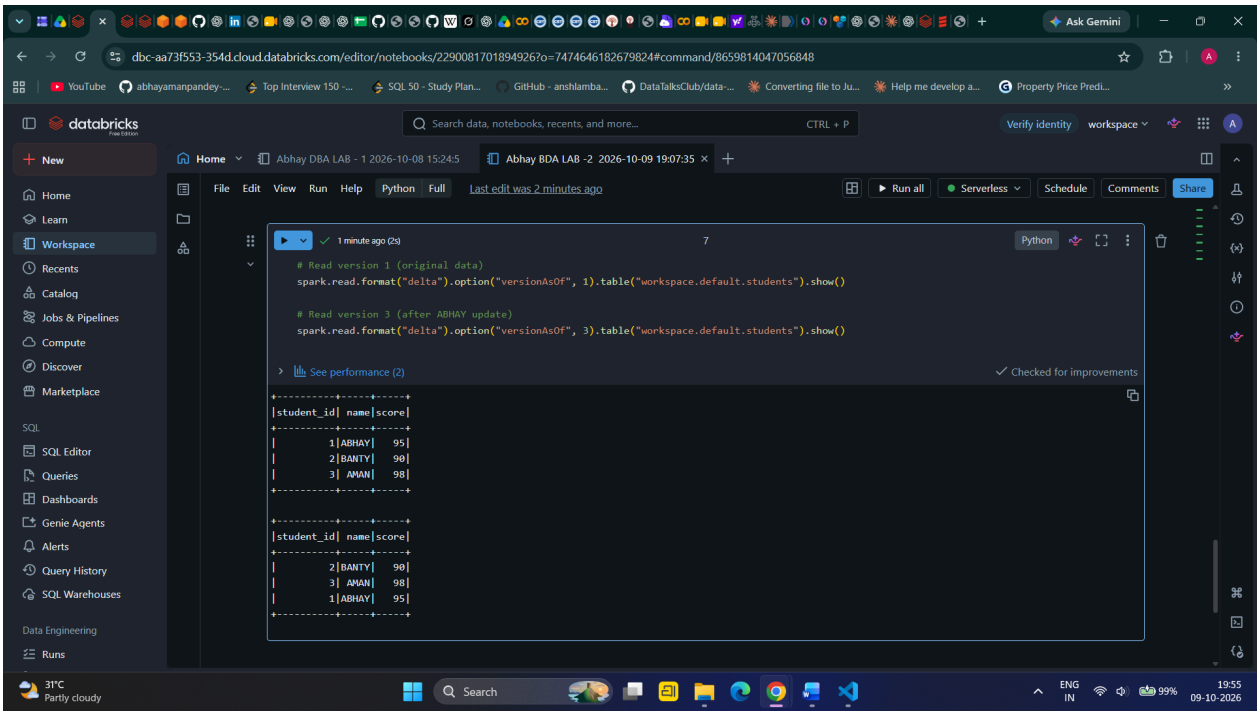
Execution screenshot 5 of 6 for Lab 2.

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LAB 2 | Delta Lake Time Travel

OUTPUT / EXECUTION EVIDENCE

Figure 2.6 - lab02-output06.png



Source: GitHub repository / Output_Lab-2

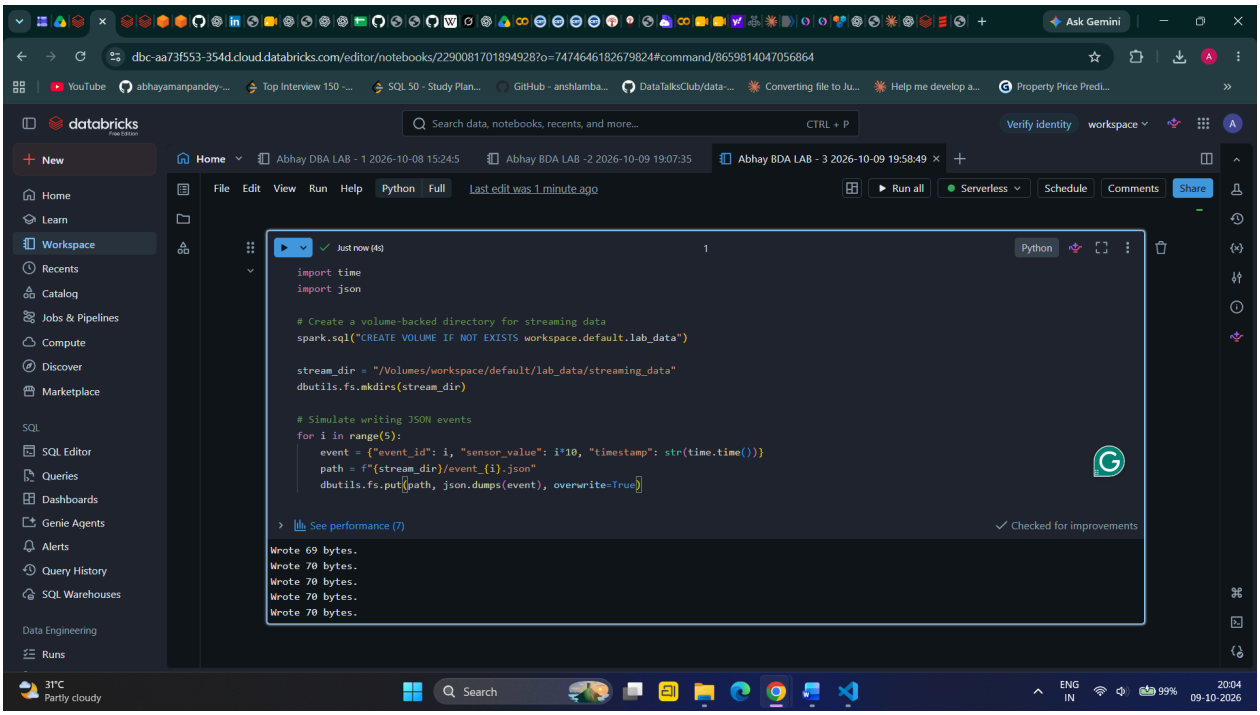
Execution screenshot 6 of 6 for Lab 2.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.1 - lab03-output01.png



Source: GitHub repository / Output_Lab-3

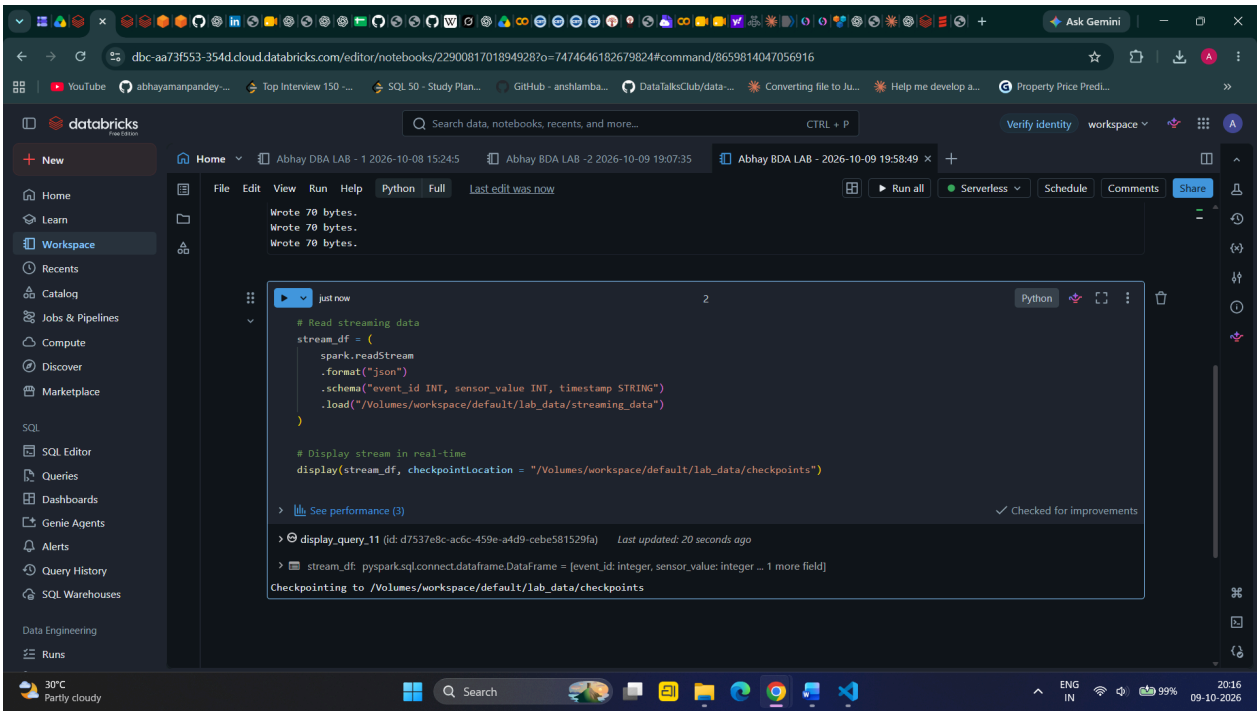
Execution screenshot 1 of 6 for Lab 3.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.2 - lab03-output02.png



Source: GitHub repository / Output_Lab-3

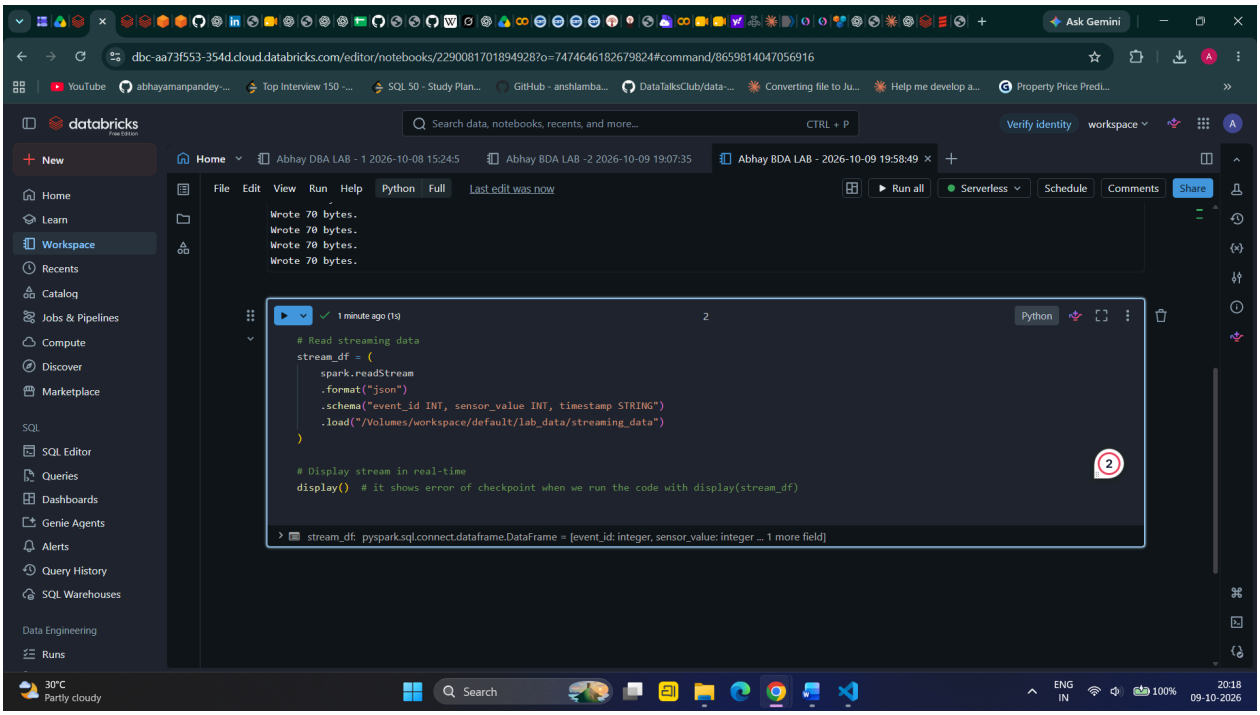
Execution screenshot 2 of 6 for Lab 3.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.3 - lab03-output03.png



Source: GitHub repository / Output_Lab-3

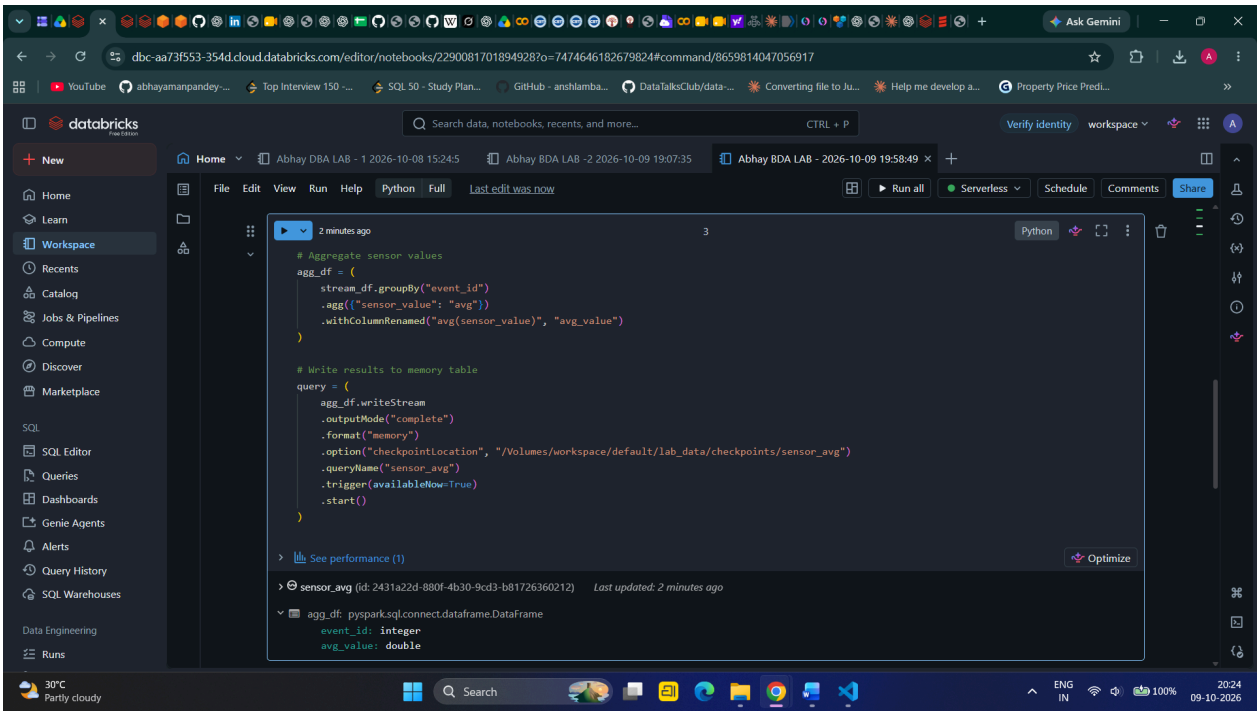
Execution screenshot 3 of 6 for Lab 3.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.4 - lab03-output04.png



Source: GitHub repository / Output_Lab-3

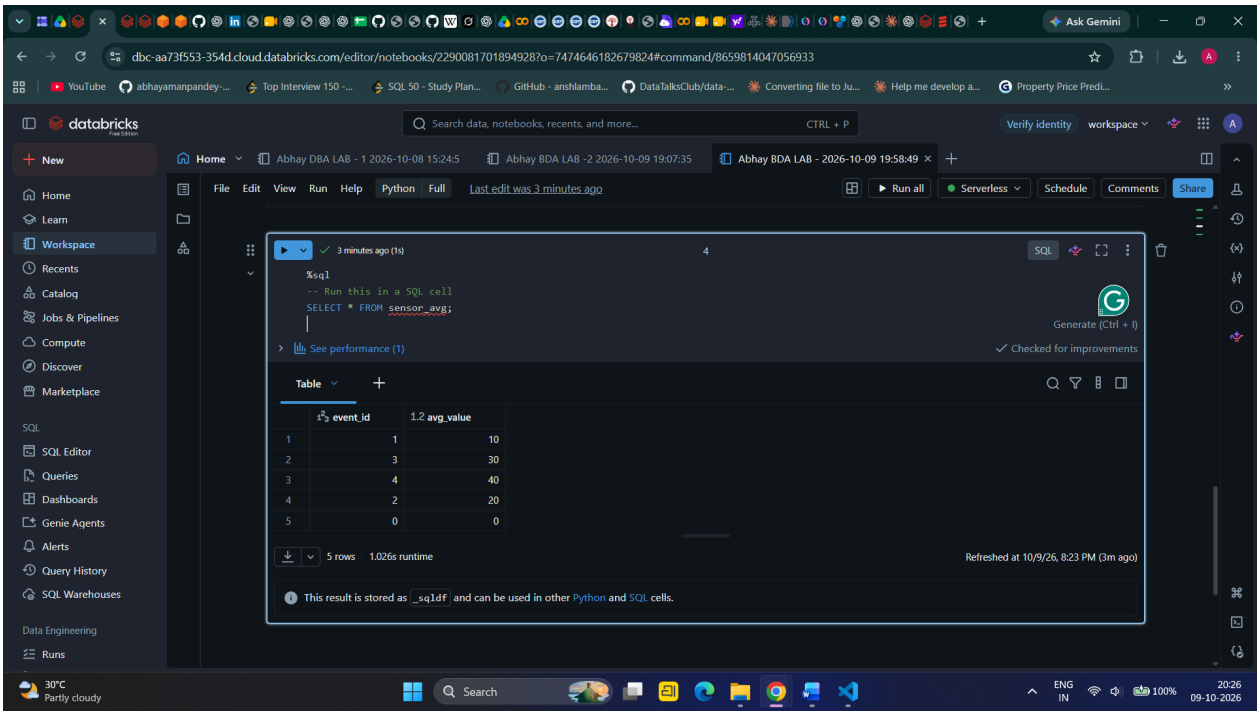
Execution screenshot 4 of 6 for Lab 3.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.5 - lab03-output05.png



Source: GitHub repository / Output_Lab-3

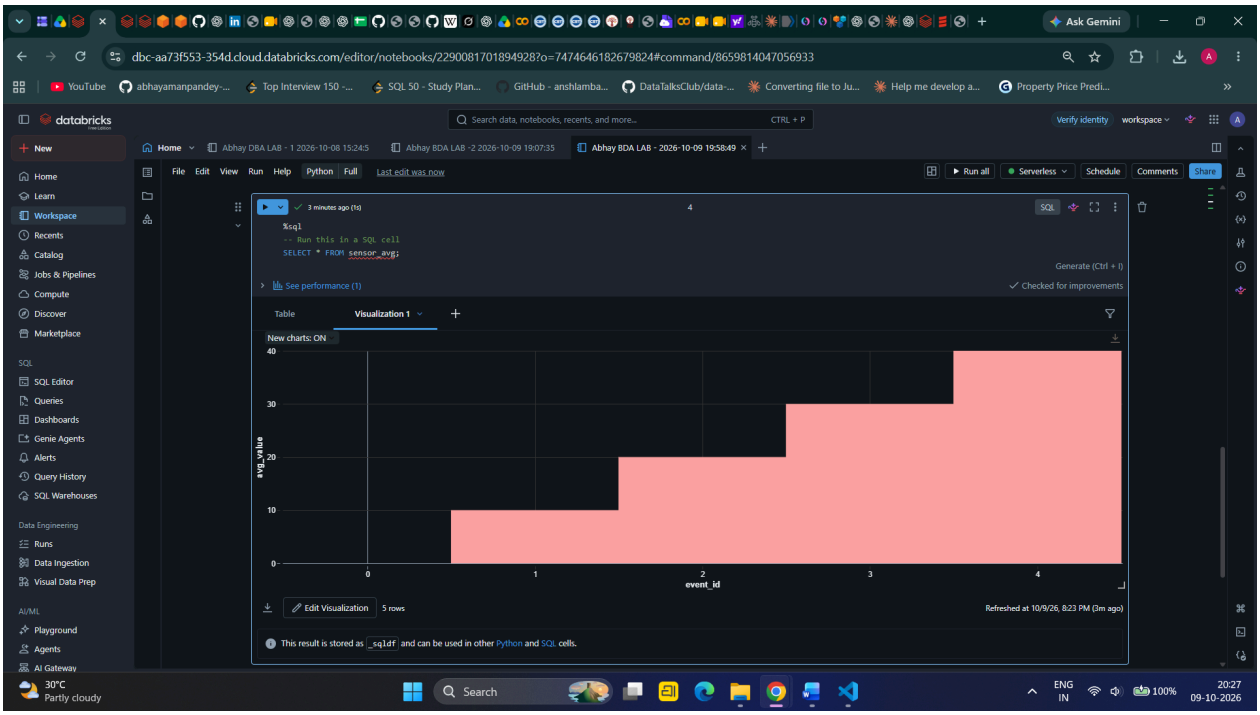
Execution screenshot 5 of 6 for Lab 3.

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LAB 3 | Streaming Data Simulation

OUTPUT / EXECUTION EVIDENCE

Figure 3.6 - lab03-output06.png



Source: GitHub repository / Output_Lab-3

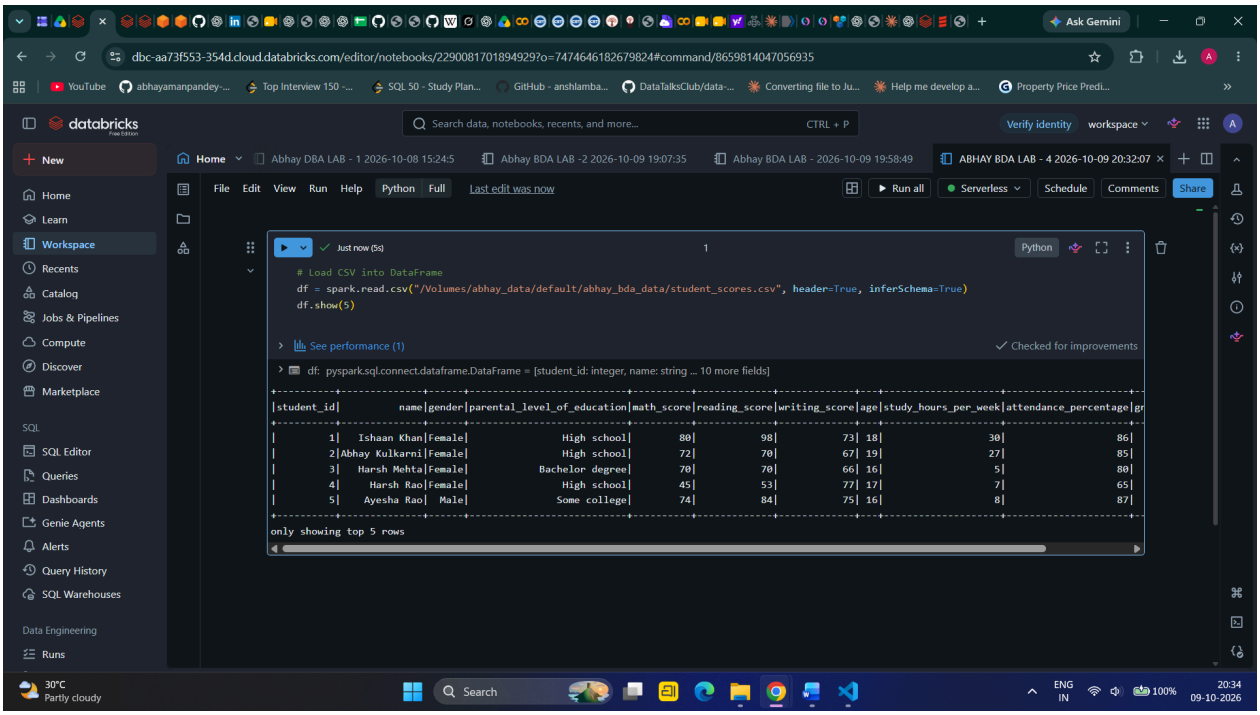
Execution screenshot 6 of 6 for Lab 3.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.1 - lab04-output01.png



Source: GitHub repository / Output_Lab-4

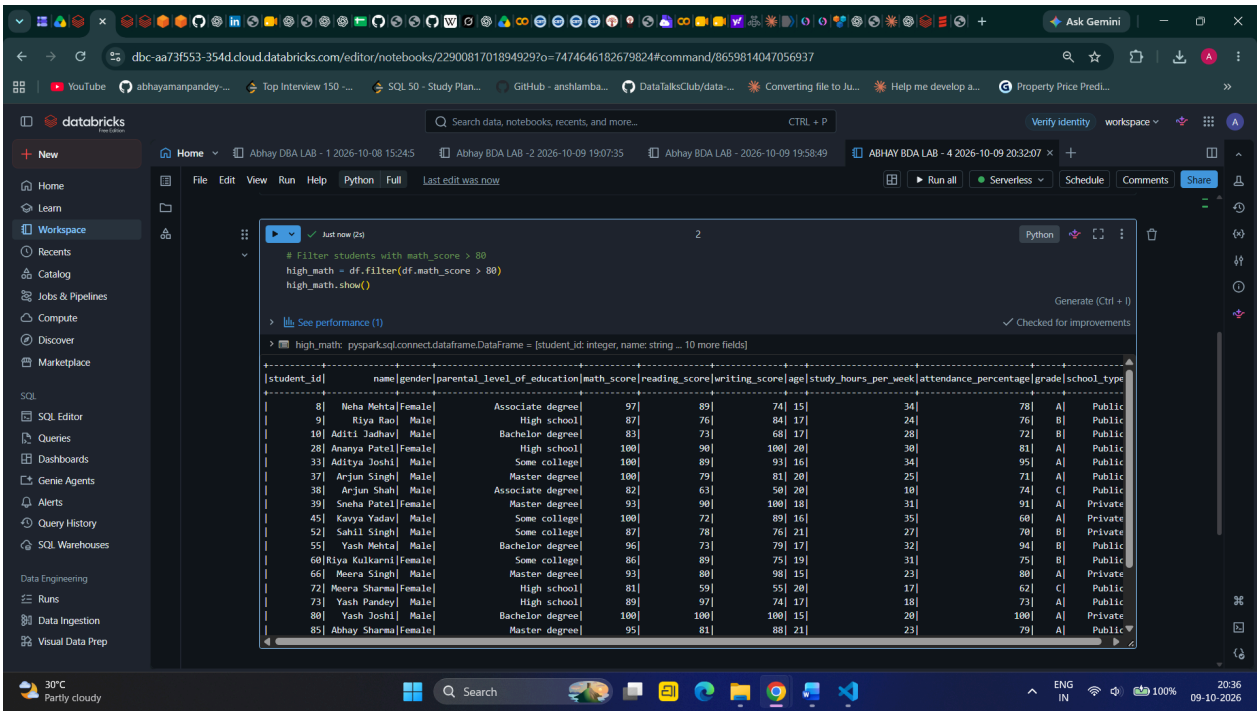
Execution screenshot 1 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.2 - lab04-output02.png



Source: GitHub repository / Output_Lab-4

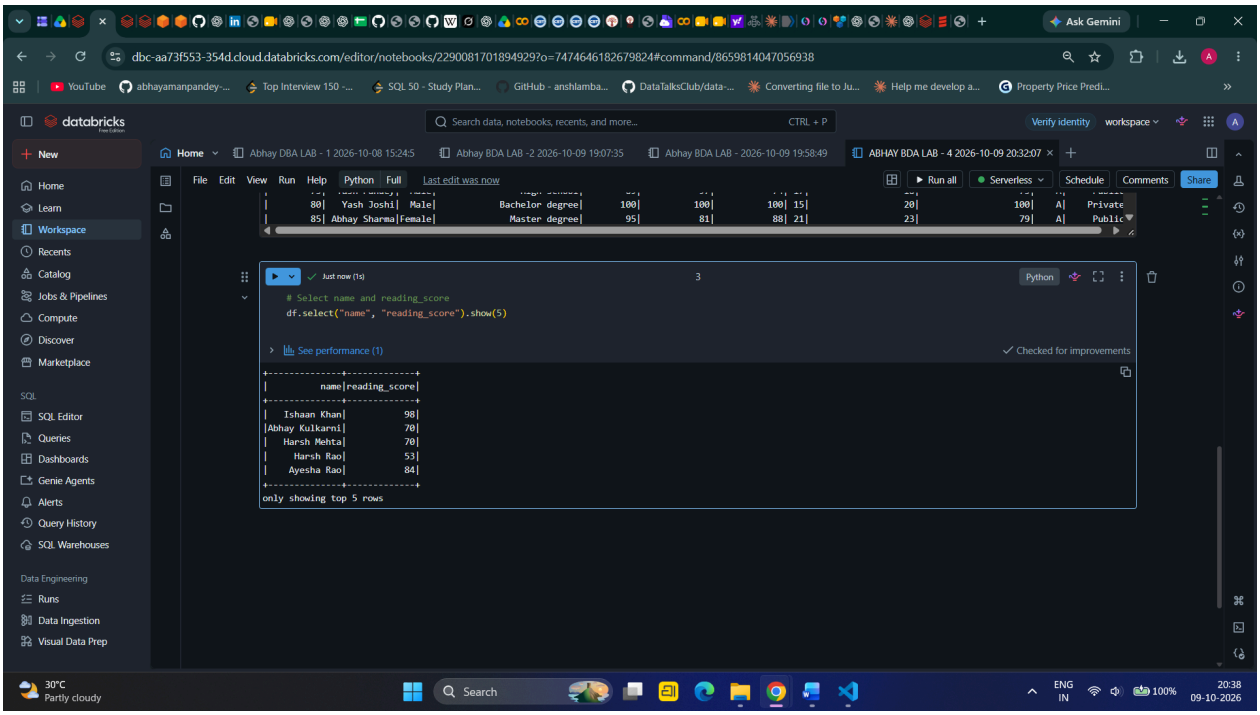
Execution screenshot 2 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.3 - lab04-output03.png



Source: GitHub repository / Output_Lab-4

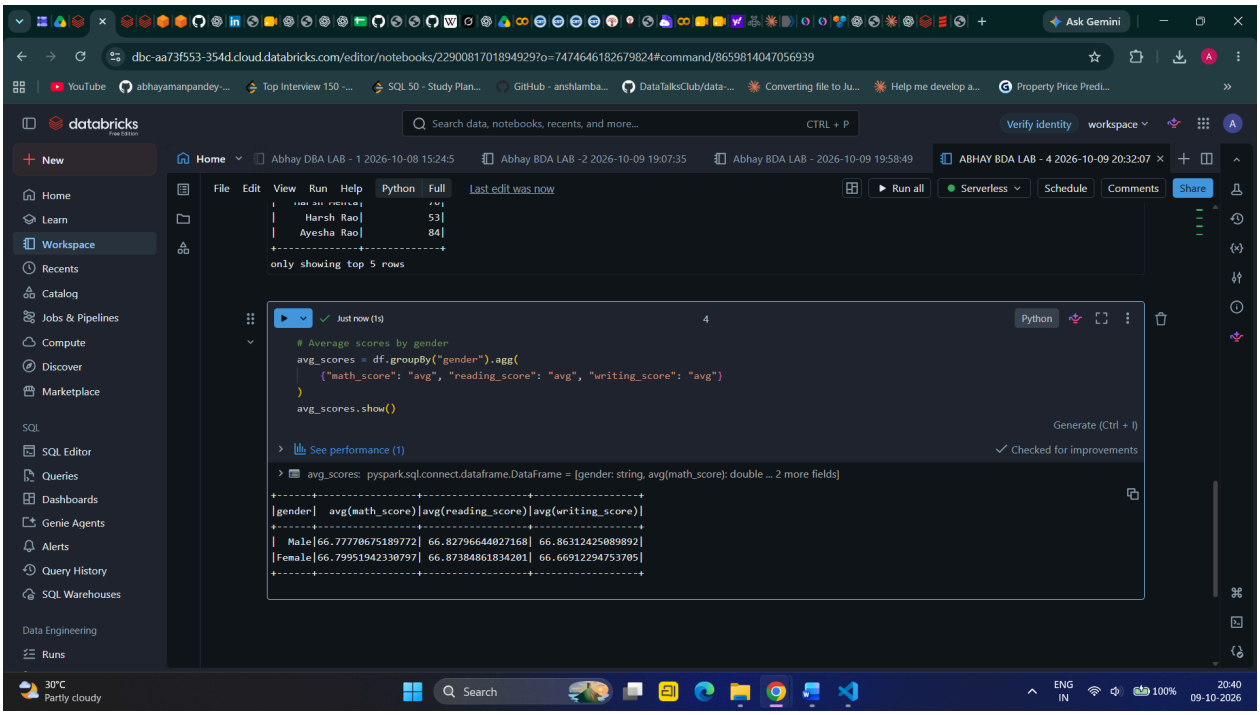
Execution screenshot 3 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.4 - lab04-output04.png



Source: GitHub repository / Output_Lab-4

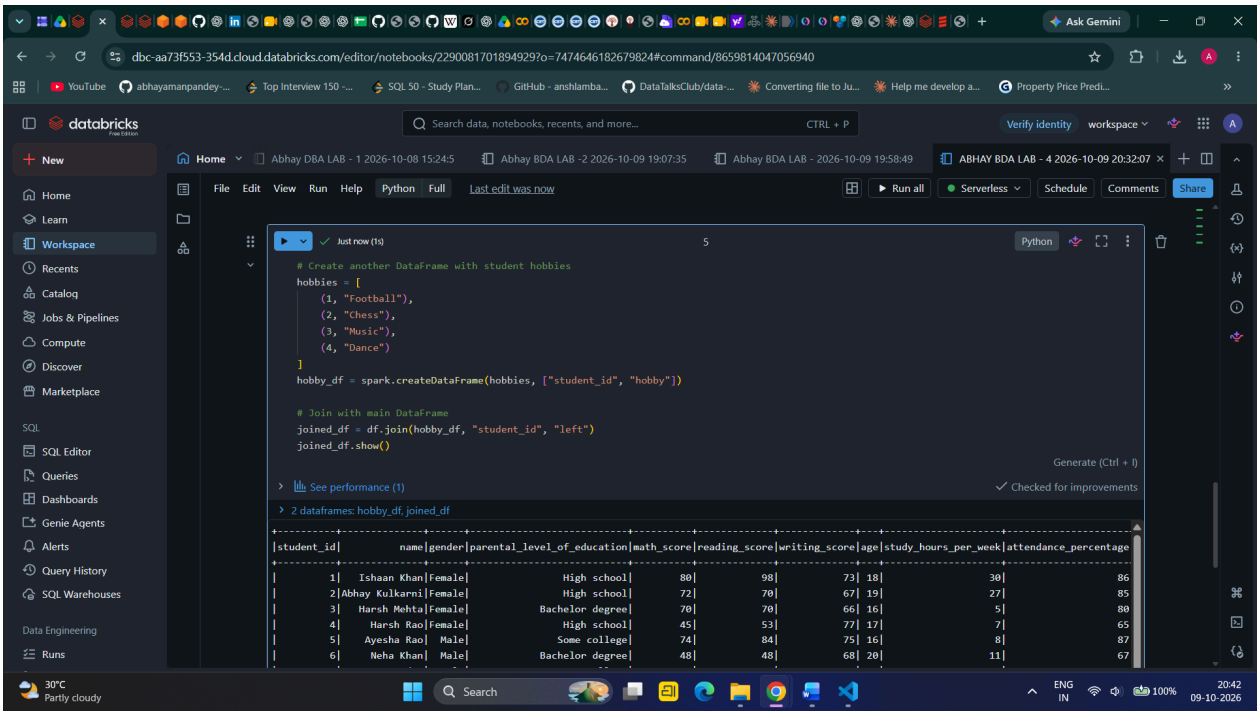
Execution screenshot 4 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.5 - lab04-output05.png



Source: GitHub repository / Output_Lab-4

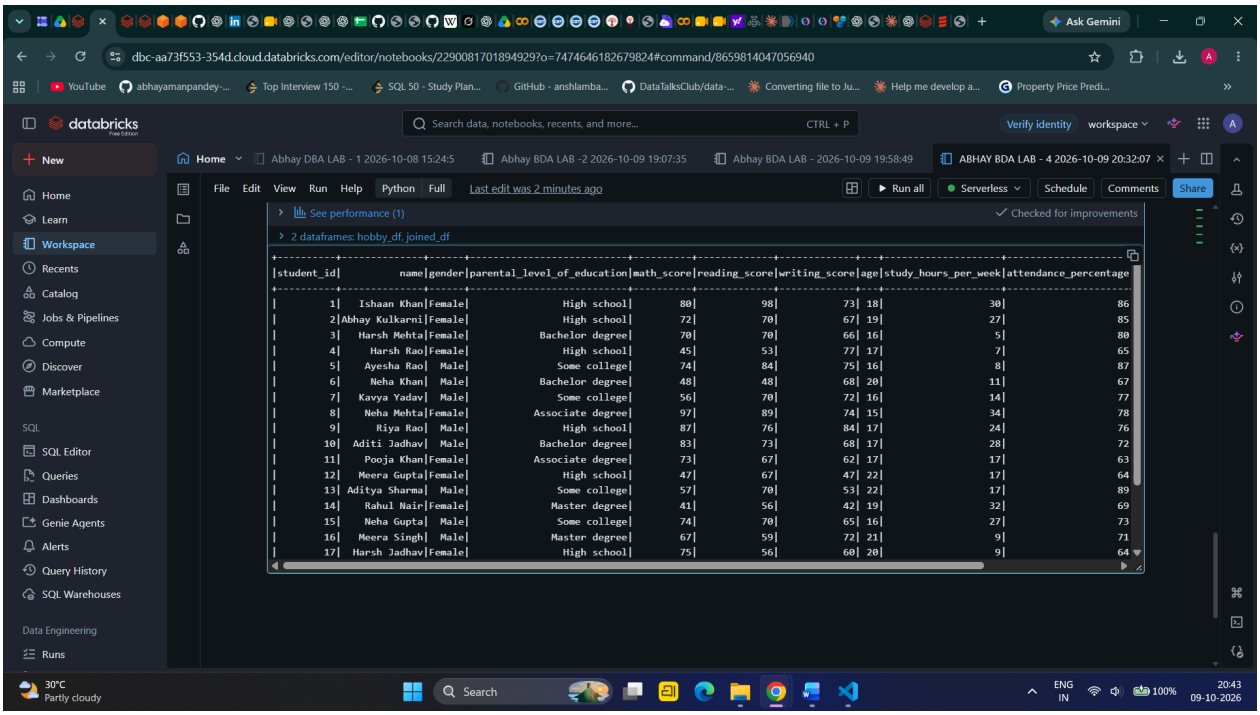
Execution screenshot 5 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.6 - lab04-output06.png



Source: GitHub repository / Output_Lab-4

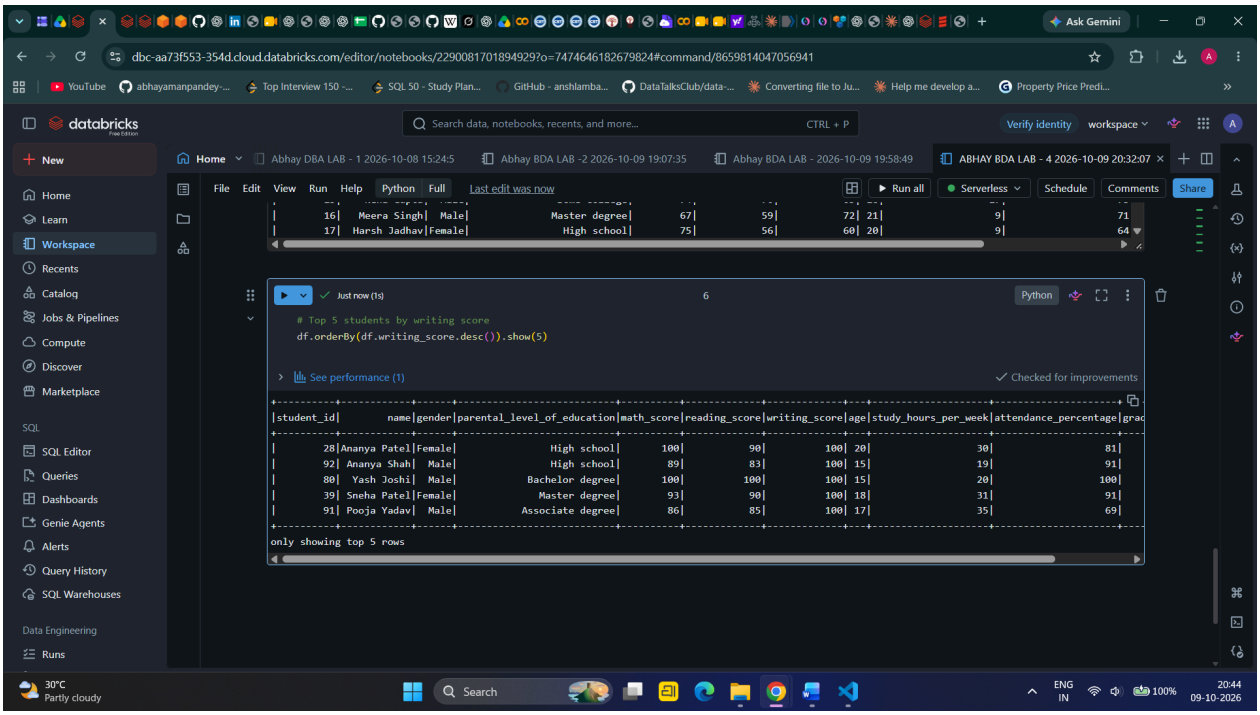
Execution screenshot 6 of 7 for Lab 4.

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LAB 4 | DataFrame Transformations

OUTPUT / EXECUTION EVIDENCE

Figure 4.7 - lab04-output07.png



Source: GitHub repository / Output_Lab-4

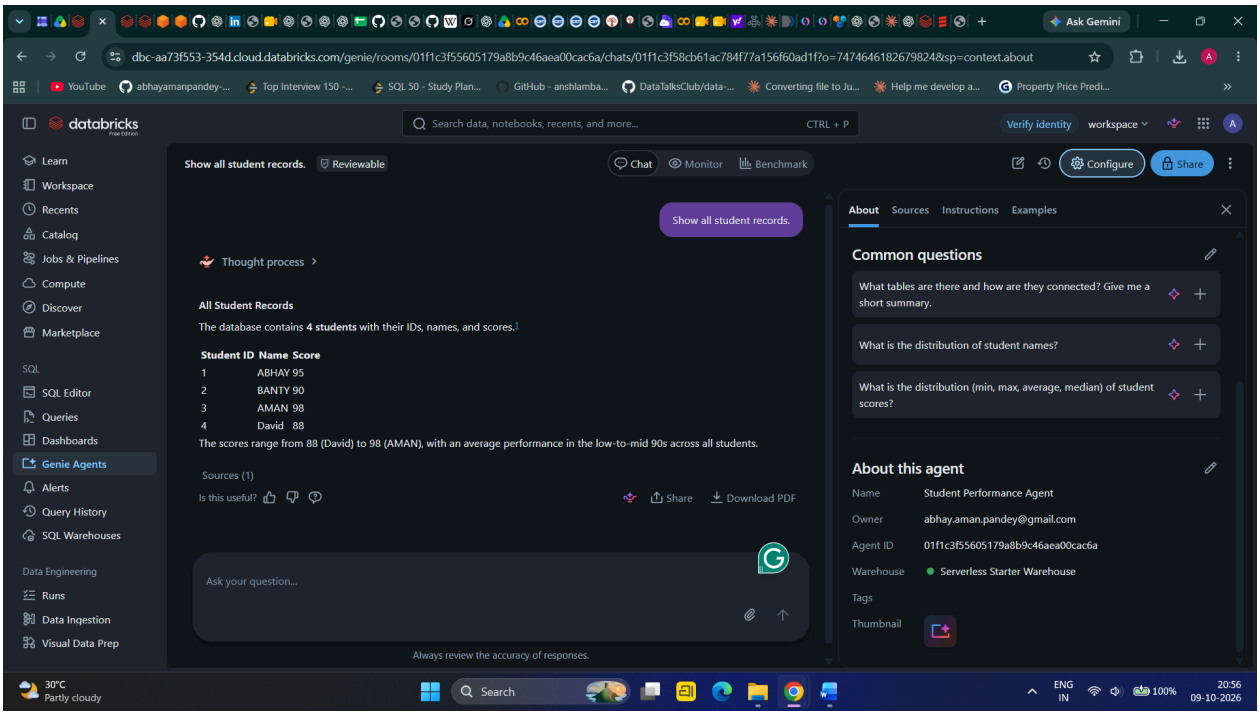
Execution screenshot 7 of 7 for Lab 4.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.1 - lab05-output01.png



Source: GitHub repository / Output_Lab-5

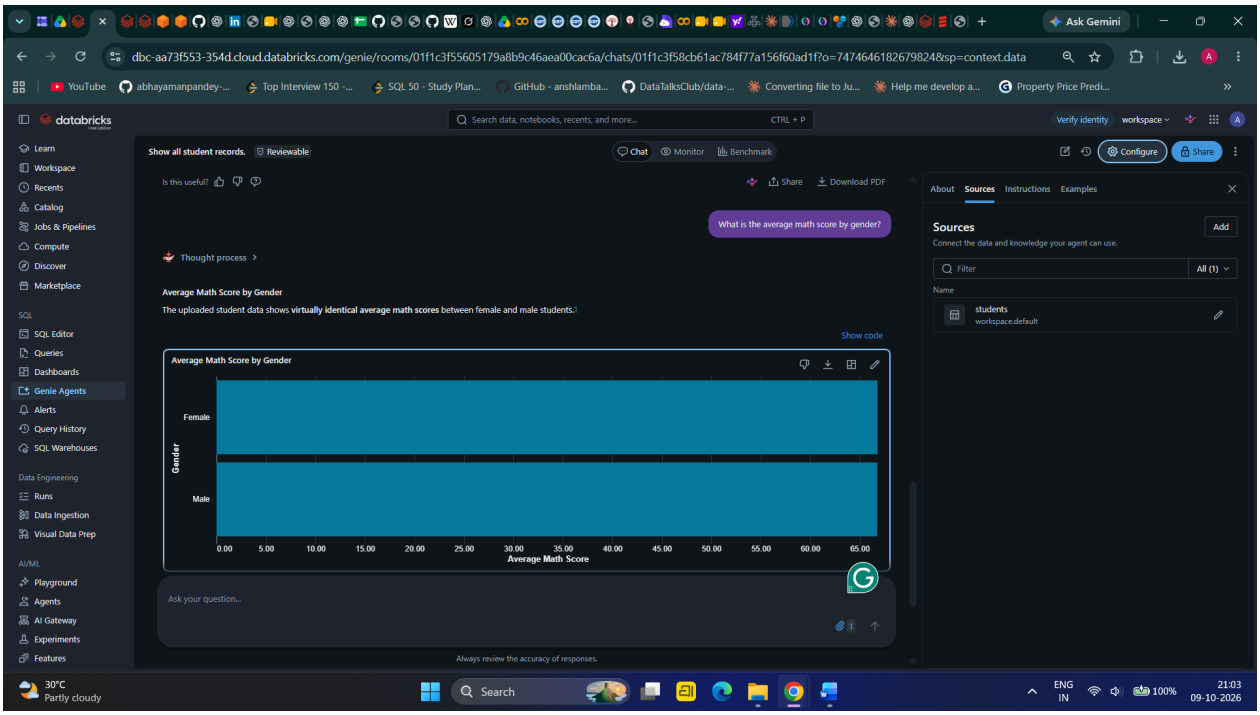
Execution screenshot 1 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.2 - lab05-output02.png



Source: GitHub repository / Output_Lab-5

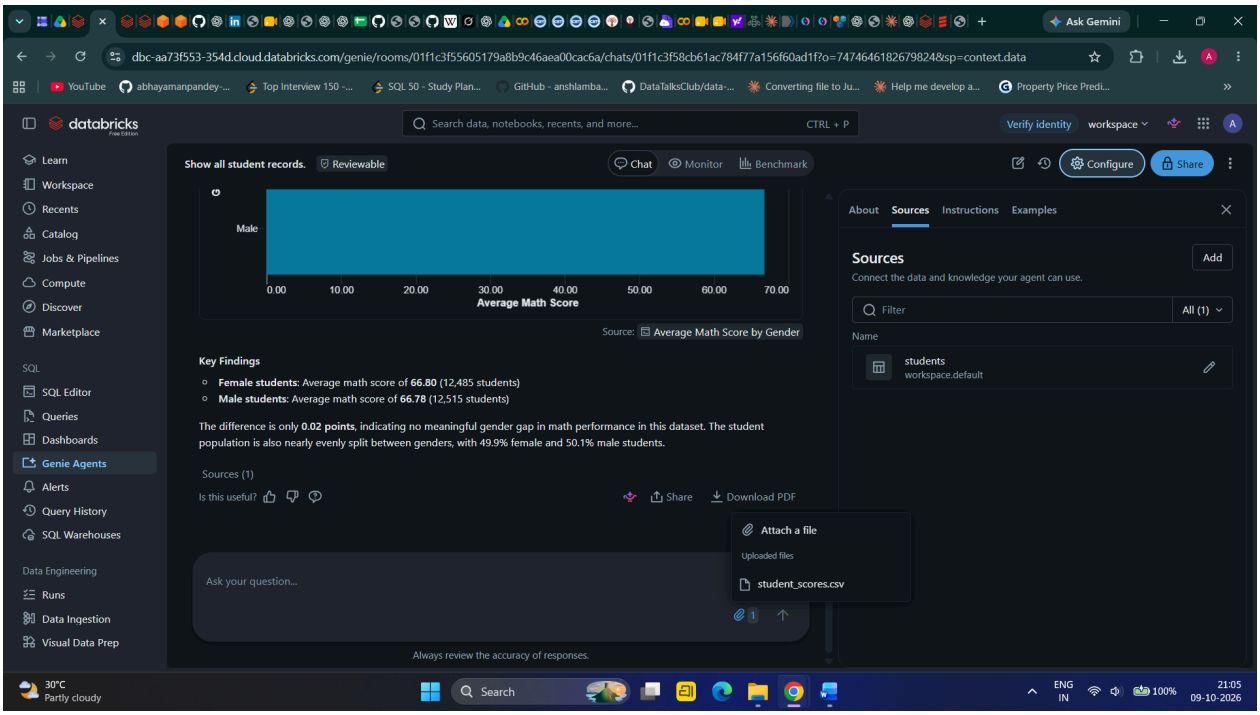
Execution screenshot 2 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.3 - lab05-output03.png



Source: GitHub repository / Output_Lab-5

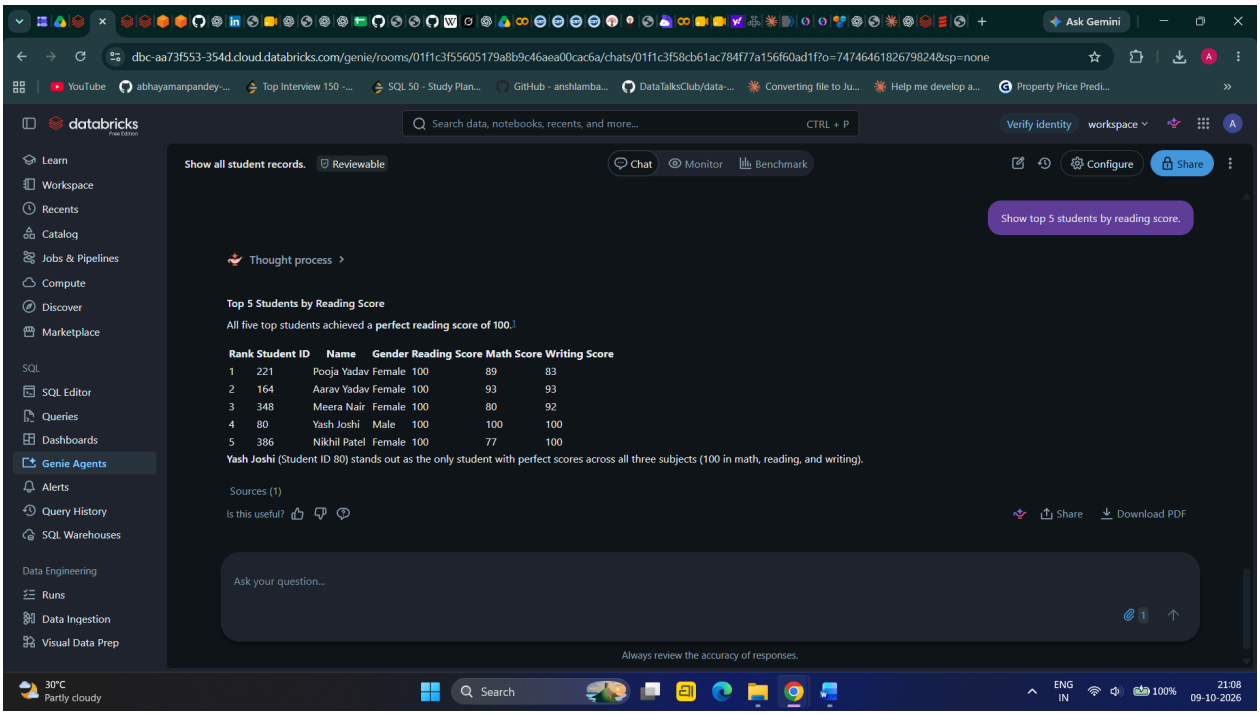
Execution screenshot 3 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.4 - lab05-output04.png



Source: GitHub repository / Output_Lab-5

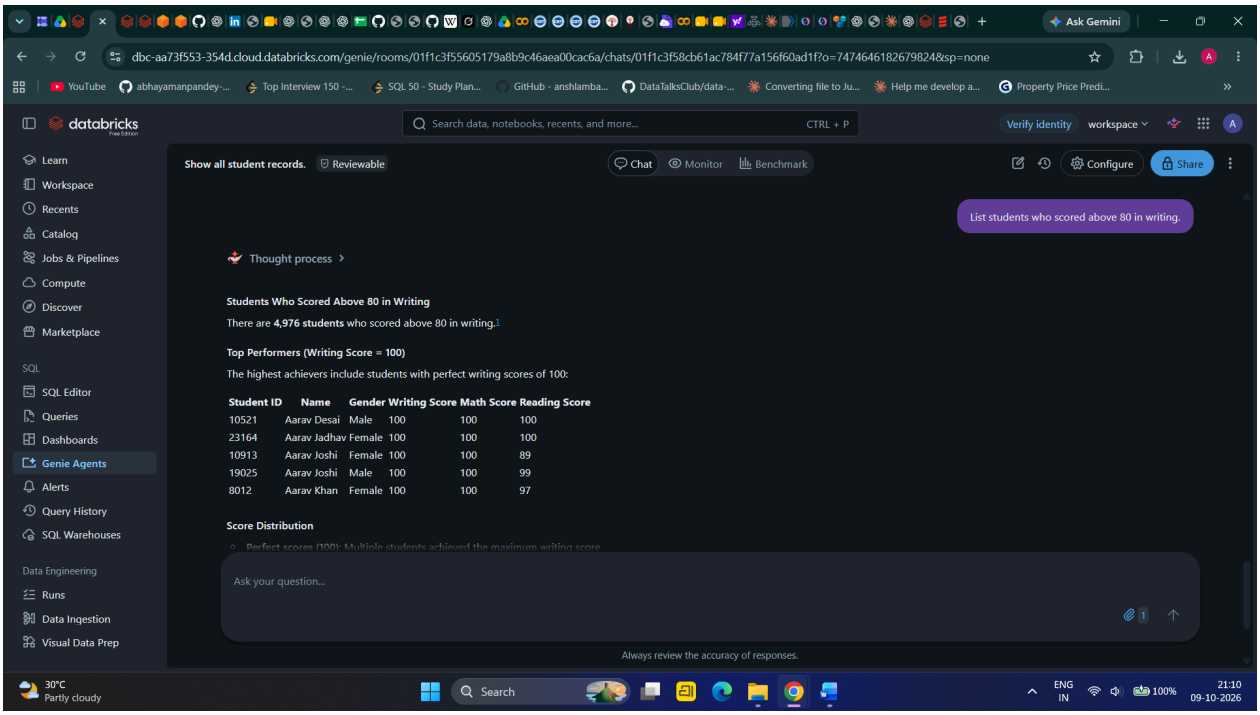
Execution screenshot 4 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.5 - lab05-output05.png



Source: GitHub repository / Output_Lab-5

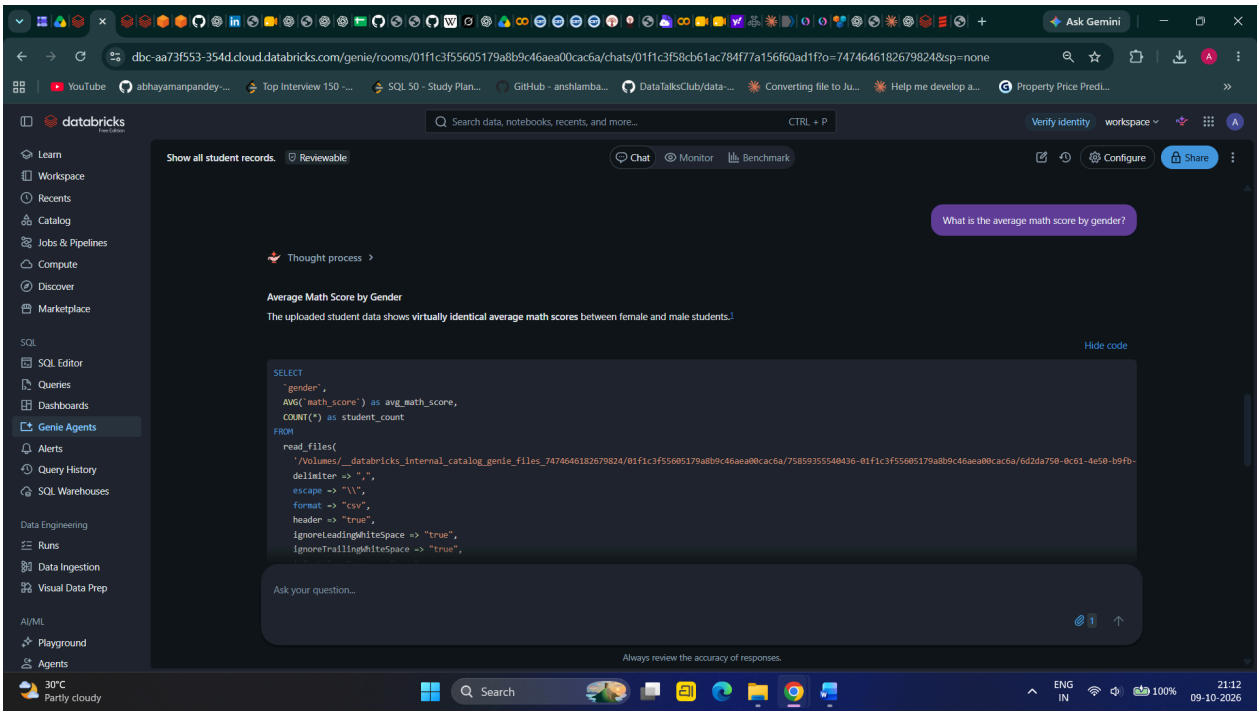
Execution screenshot 5 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.6 - lab05-output06.png



Source: GitHub repository / Output_Lab-5

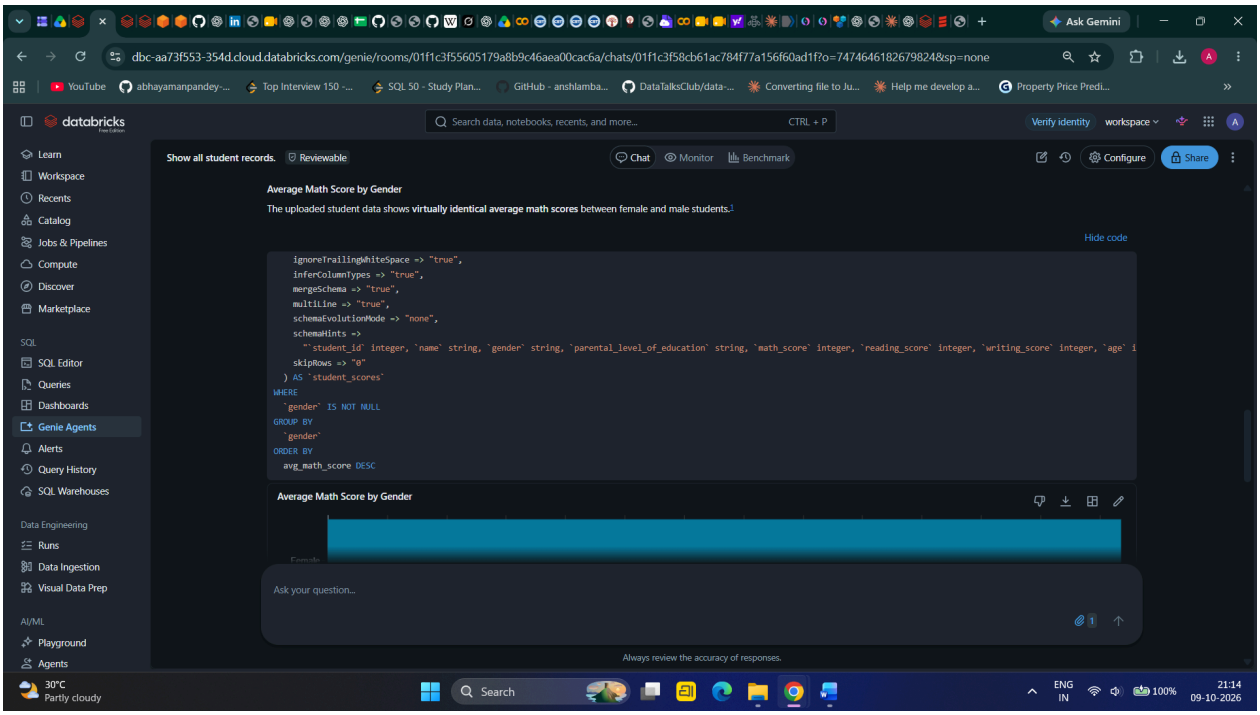
Execution screenshot 6 of 7 for Lab 5.

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LAB 5 | Genie Agent Demo

OUTPUT / EXECUTION EVIDENCE

Figure 5.7 - lab05-output07.png



Source: GitHub repository / Output_Lab-5

Execution screenshot 7 of 7 for Lab 5.

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CONCLUSION

Lab 1 - Spark SQL Basics

In this experiment, I worked with CSV datasets in Databricks and practiced loading records into Spark DataFrames, creating SQL views, and querying data using SELECT, GROUP BY and ORDER BY. The exercise helped me understand how Spark SQL supports familiar SQL operations on distributed data.

Lab 2 - Delta Lake Time Travel

I learned how Delta tables maintain transaction history while data is updated or appended. By examining table versions and reading previous states, I understood the practical purpose of Delta Lake time travel, version control and ACID-style reliability.

Lab 3 - Streaming Data Simulation

I practiced processing simulated JSON sensor events using Spark Structured Streaming. The experiment demonstrated how a file-based stream can be continuously read, aggregated and queried, highlighting the difference between streaming and conventional batch processing.

Lab 4 - DataFrame Transformations

I applied filtering, column selection, grouping, aggregation, joins and sorting using Spark DataFrames. This lab improved my understanding of how transformations can be combined to prepare and analyze structured datasets in Databricks.

Lab 5 - Genie Agent Demo

I explored how Databricks Genie supports natural-language questions about cataloged datasets and converts analytical questions into SQL-backed results. The experiment illustrated how conversational analytics can make data exploration easier for users without writing every query manually.

Repository: github.com/abhayamanpandey-max/BDA-Databricks-Assignment